NOT MEASUREMENT SENSITIVE

MIL-T-83823 (USAF)
13 September 1991

MILITARY SPECIFICATION

TRUCK, FIRE FIGHTING (STRUCTURAL OR BRUSH-STRUCTURAL, 1,250 GPM, 4 X 2 OR 4 X 4 CHASSIS)

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers a diesel engine-driven four wheel, two wheel drive (4X2), 1,250 GPM pumping capacity, structural fire fighting truck and a diesel engine-driven four wheel, four wheel drive (4X4), 1,250 GPM pumping capacity, brush-structural fire fighting truck. See paragraph 6.2 for type vehicles to be furnished.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification for the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-S-300	Sheeting and Tape, Reflective, Nonexposed Lens.
W-B-131	Battery, Storage (Vehicular, Ignition Lighting and Starting)
QQ-P-416	Plating, Cadmium
VV-F-800	Oil, Diesel
ZZ-H-451	Hose, Fire, Woven Jacketed, Rubber or Latex, or Rubber Coated Fabric Lined, with Couplings
ZZ-H-561	Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water
GGG-A-926	Ax, (Single Bit, Double Bit, Pick Head and Mattock Head)
GGG-W-665	Wrench, Spanner
KKK-T-2108	Trucks and Truck Tractors; Commercial, Diesel or Gasoline Engine Driven, 24,000 to 32,000 Pounds GVW, 4X2

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/LZDTB, Robins AFB GA 31098-5365, by using the Standardization Document Improvment Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4210

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

KKK-T-2110	Trucks and Truck Tractors, Commercial, Diesel or Gasoline Engine Driven, 16,000 to 34,000 Pounds GVW, 4X4

MILITARY

MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-H-4497	Hose, Cotton, Rubber Lined, Water, Braided Construction with Couplings
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements For
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-S-11086	Strap, Hose, and Ladder
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-F-24385	Fire Extinguishing Agent, Aqueous Film Forming Foam (AFFF) Liquid Concentrate, for Fresh and Sea Water
MIL-T-25959	Tie Downs, Cargo, Aircraft
MIL-P-26915	Primer Coating, Zinc Dust Pigmented, for Steel Surfaces
MIL-T-27260	Tie Downs, Cargo, Aircraft, CGU-1/B
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-C-52404	Connections, Hose, Fire and Water
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component Chemical Agent Resistant
MIL-S-81733	Sealing and Coating Compound, Corrosion Inhibiting
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate for Aerospace Applications
MIL-N-87963	Nozzle: Fuel, Aircraft, Over-wing Type MD-3

STANDARDS

FEDERAL

FED-STD-297D	Rustproofing of Commercial (Nontactical) Vehicles
FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations for use on Drawings, Specifications, Standards, and Technical Documents
MIL-STD-17	Mechanical Symbols for Aeronautical, Aerospacecraft and Spacecraft use
DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

MIL-STD-281	Automobile, Trucks, Truck Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-454	Standard of General Requirements for Electric Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-721	Definition of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-808	Finishes, Materials and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-965	Parts Control Program
MIL-STD-970	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-1250	Corrosion Prevention and Deterioration Control In Electronic Controls and Equipment
MS51317	Light, Warning, Vehicular, Rotating, D.C.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities
MIL-STD-1556	Government/Industry Data Exchange Program
MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems
MIL-STD-1587	Materials and Process Requirements for Air Force Weapons Systems
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

HANDBOOKS

MIL-HDBK-781	Reliability Design Qualification Production Acceptance Tests: Exponential Distribution

NATIONAL BUREAU of STANDARDS

Handbook H28	Screw Thread Standards for Federal Services

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

LAWS AND REGULATIONS

49CFR393	Federal Motor Carrier Safety Regulations
49CFR571	Federal Motor Vehicle Safety Standards

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the currentversion before use.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for the issuance thereof.)

DRAWINGS

AIR FORCE

8838302	Fastener Quality Assurance Requirements
AF FORM 1812	Operator's Inspection Guide and Trouble Report

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)

EUROPEAN TYRE AND RIM TECHNICAL ORGANIZATION

Standards

(Application for copies should be addressed to the European Tyre and Rim Organization, 32 Avenue Brugmann-B, 1060,Brussels Belgium.) New York, NY 10017.

Tire and Rim Association, Inc.

Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Incorporated, 3200 W. Market Street, Akron OH 44313).

National Fire Protection Association (NFPA)

NFPA 1500	Fire Department Occupational Safety and Health Program
NFPA 1901	Automotive Fire Apparatus
NFPA 1931	Fire Department Ground Ladders

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Quincy, MA 02269).

Society of Automotive Engineers, Incorporated

SAE Standards and Recommended Practices

J185	Access Systems for Off-Road Machines
J198	Windshield Wiper Systems, Trucks, Busses and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

	Multipurpose Vehicles
J293	Vehicle Grade Parking Performance Requirements
J336a	Sound Level for Truck Interior
J360	Motor Vehicle Grade Parking Performance Test Code
J366	Exterior Sound Level for Heavy Trucks and Buses
J534	Standard for Lubrication Fittings
J537	Storage Batteries, Standard
J706	Rating of Winches
J833	Male Physical Dimensions for Construction and Industrial Equipment Design
J925	Minimum Service Access Dimensions for Off-Road Machines
J994	Alarm Back Up Electric Performance, Test and Application
J1050a	Describing and Measuring the Driver's Field of View
J1349	Engine Test Code - Spark Ignition and Diesel

(Application for copies should be addressed to Customer Service Department, Publications Group, SAE, 400 Commonwealth Drive, Warrendale, PA 15096

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.1.1.1 Parts control. The contractor shall participate in Government Industry Data Exchange Program (GIDEP) in accordance with MIL-STD-1556B. In the design of the vehicle, the contractor shall ensure the components selected have not been identified as a problem or unresolved deficiency by GIDEP. The contractor shall submit an Alert/Safe-Alert, on a DD Form 1939, whenever an actual or potential problem is evident with respect to a part, component, material, test equipment, process or safety condition. The contractor shall certify maximum use of off-the-shelf items in accordance with MIL-STD-965A.

3.1.2 Materials. The selection of materials and processes shall be in accordance with MIL-STD-1568 and MIL-STD-1587. Materials not covered by this specification or referenced documents shall be suitable for this type of application and the intended environment. Alternate materials, such as composites, polymers, etc., may be used if suitable for the intended use and environmental extremes. The contractor shall not use nonstandard or proprietary materials and processes without prior government approval. All commercial equipment and parts shall comply with the vehicle life expectancy requirements and shall be suitable for the intended use and environmental extremes.

3.1.2.2 Prohibited materials. Magnesium and magnesium alloys shall be subject

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

to approval by the Government. Wood, wood products, polyvinylchloride, polyester and corrosive RTV (yielding acetic acid upon curing) are prohibited for use in all forms for the interior and exterior of this vehicle. Asbestos in all forms, including brake linings and gaskets, shall not be used.

3.1.2.3 Metals. Unless otherwise specified herein, parts and components of the truck which come in contact with fire fighting agents and which come in contact with coolant liquid, except the engine, shall be fabricated of material which can withstand exposure to these agents for 12 years without penetrating or deteriorating corrosion.

3.1.2.4 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.2.5 Protective treatment. When materials are used that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used.

3.1.3 Design and construction. To the greatest extent, parts and components between the 4X2 and 4X4 models shall be interchangeable. The truck shall be so constructed that parts do not work loose in service and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping, and storage with minimum loss of time for maintenance, repair and servicing. All installed handles, latches, controls, mounted equipment, compartments and access panels shall be arranged and located so that no interferences occur during operations, and shall be readily accessible by a person wearing arctic clothing and mittens, as defined by SAE J185 and J925, for operation, removal, replacement and maintenance. Structures shall be designed to avoid pockets where debris or moisture can accumulate. Complete drainage shall be provided from the lowest points of closed structural configurations, and both internal and external pockets and areas. Joint design shall minimize crevices formed by overlapping surfaces and shall point away from the forward direction of travel or point down. Lap joints shall be coated with sealant before assembly and shall have no gaps. Where double panel construction is used that does not require service access the panels shall be foam filled with non-reverting polyurethane.

3.1.3.1 Functional design. The truck shall be designed to operate efficiently at military installations world wide for the purpose of providing quick response for suppression and extinguishment of real property and material fires as specified herein under extreme climatic conditions. The truck shall be capable of operating on:

3.1.3.1.1 Four by two, structural truck. Paved surfaces and unimproved roads with ice, snow, or muddy surfaces as required to accomplish the fire fighting mission at such installations.

3.1.3.1.2 Four by four, structural truck. Paved surfaces, unimproved roads with ice, snow, or muddy surfaces and on cross country terrain as required to accomplish the fire fighting mission at such installations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.1.3.2 Reliability. The contractor shall establish, conduct, and maintain a comprehensive reliability program that satisfies the following requirements of MIL-STD-785: tasks 101, 103, 104, 105, 204, and 303. The vehicle, including all components, shall have lower test mean time between failure (MTBF) of not less than 200 hours, demonstrated in accordance with Test Plan XVIID of MIL-HDBK-781 and a lower test mean time between critical failure (MTBCF) of not less than 400 hours of operation, demonstrated in accordance with Test Plan XXID of MIL-HDBK-781.

A failure is defined as the event or inoperable state in which any item or part of an item does not, or would not, perform as previously specified. A critical failure is defined as a failure or combination of failures that prevents the vehicle from performing its specified mission.

Mean repair time (MRT) is the elapsed time to accomplish all repairs divided by the number of repairs performed. Mathematically, MRT is expressed as:

$$MRT = T/N,$$

where T is Elapsed Time
N is Number of Repairs.

The elapsed time includes all maintenance actions required to correct the malfunction, including time to prepare, troubleshoot, remove, replace, repair, adjust, and check. MRT does not include administrative or supply delays.

Definitions of reliability terms not included herein shall be as found in MIL-STD-721. The vehicle shall be designed for a life expectancy of at least 15 years.

3.1.3.3 Maintainability. Accomplishment of all servicing tasks shall be by personnel possessing Air Force Specialty Code (AFSC) 57150. Accomplishment of all scheduled preventive maintenance, excluding major overhaul, and not less than 98 percent of all corrective maintenance tasks shall be by personnel possessing AFSC 47251A. The truck shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with performances required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the truck. Parts and components shall be located for rapid inspection and recognition of excessive wear or potential failure. Service access dimensions shall be in accordance with SAE J925, for servicing while wearing arctic clothing and mittens.

c. Ease of adjusting, servicing, and replacing parts and components.

d. Use or readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

f. Mean corrective maintenance downtime shall be no greater than one hour.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the currentversion before use.

g. Maximum corrective maintenance downtime shall be no greater than four hours.

h. Maximum total time to perform all daily scheduled inspection tasks listed on AF Form 1812, Operator's Inspection Guide and Trouble Report, shall be no greater than 35 minutes for one person.

3.1.3.3.1 Maintenance tasks. The truck shall be designed so that one mechanic, using common hand tools, can perform the following maintenance operations within the time specified:

(1) Remove, replace and adjust all engine drive belts within 45 minutes.

(2) Remove and replace the alternator(s) within 30 minutes.

(3) Remove and replace engine coolant and system hoses, except heater hoses, within 45 minutes.

(4) Drain engine oil, remove and replace oil filter element(s) and refill crankcase within one hour.

(5) Remove and replace fuel filter element(s) within 30 minutes.

(6) Disconnect battery cables, remove and replace batteries and reconnect cables within 30 minutes.

(7) Drain transmission oil, remove and replace all filter element(s) and strainer(s) and refill within one hour.

(8) Remove and replace starter within one hour.

(9) Remove or install the engine or transmission/transfer case, using two mechanics within four hours.

3.1.4 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an asymmetric mounting system shall be provided to ensure proper orientation. If an asymmetric mounting means is not practical, the item shall be so designed that there is only one proper operating position.

3.1.5 Major assemblies. Major assemblies shall be removable without the necessity for draining liquids or coolants, or dismantling other assemblies. Major assemblies shall be equipped with lifting eyes or other provisions for attachment of lifting facilities.

3.1.6 Access Covers. Access covers, which must be removed for component adjustment, inspection or parts removal, shall be equipped with quick disconnect fasteners. Fasteners shall be retained to either the truck or the access cover to prevent loss.

3.1.7 Fastening devices. Ferrous alloy fasteners shall be cadmium plated per QQ-P-416, Type II, to a minimum thickness of 0.0003 inches unless thread tolerances dictate otherwise. Aluminum alloy fasteners shall be anodized per

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

MIL-A-8625. Fastener installation shall be in accordance with MIL-STD-1568. All screws, bolts, nuts pins and similar fasteners shall be installed with an adequate means for preventing loss of proper torque or adjustment. When subject to removal, replacement, or adjustment, attaching hardware for any component shall be easily removable and shall not be swaged, peened or staked. Tapped aluminum threads shall have a minimum threaded engagement of two times the nominal fastener diameter. All fasteners shall follow the quality assurance requirements in Air Force Drawing 8838302. Adhesive applied wiring supports shall not be used.

3.1.7.1 Permanent fasteners. All riveted joints and mating/faying surfaces of exterior and interior components shall be sealed using sealing compound in accordance with MIL-S-81733.

3.1.8 Servicing provisions. Service drains, filler plugs, grease fittings, bleeders, and checkpoints for all components shall be accessible and shall not require special tools.

3.1.9 Safety Features. All occupied spaces or space in which work is performed during operation, service, and maintenance of the truck shall be free of hazardous protrusions, sharp edges, cracks, or other features that may cause injury to personnel. Safety features such as anti-skid steps, handrails, or guards shall be provided at all points where the protection of personnel is in question. All critical switches shall be covered and located so that they are protected from being operated accidentally. Electrical switches/components shall be weatherproof when exposed to outside conditions. All grab handles and handrails shall be fabricated of 1-1/4 inch rubber covered tubing or of 1-1/4 inch, polished or plated, tubing with a knurled surface. The grab handle tubing and standoff brackets shall be of adequate wall thickness and strength to support the dynamic loads of a 275 pound firefighter, including gear, generated in emergency operation.

3.1.9.1 Hazard guards. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized shall be insulated or guarded to the extent necessary to eliminate the hazard.

3.1.9.2 Component protection. All hydraulic lines, air system tubing, control cables and all electrical lines shall be clipped to the frame or body structure of the truck and, except where a through-frame connector is necessary, shall be furnished with metal protective looms or grommets at each point where they pass through body panels or structural members.

3.1.9.3 Brush Guards (4X4 trucks only). The radiator, engine oil pan, transmission oil pan and transfer case shall be located or protected by structural guards to prevent damage to the components from brush, stones, logs, et cetera, likely to be encountered by the truck during cross-country operation. Brush guards shall drain completely. Each end of the drive shafts shall be guarded to prevent dropping on the ground should either or both universal joints fail. All electrical switches on the exterior of the cab or body, including the pump operator's control panel, shall have safety covers to prevent accidental activation by trees or branches likely to be encountered during cross country operation. Front and rear warning lights, cab directional lights and air horns shall be protected with removable tubular guards. The muffler and tailpipe

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

shall be protected from contact with the ground.

3.1.10 Lubrication. All parts of the truck requiring lubrication shall be provided with lubrication fittings conforming to SAE J534. Where the use of high lubricating pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication.

3.1.11 Human engineering. Human engineering design criteria for fifth percentile female and ninety-fifth percentile male in MIL-STD-1472 shall be applied to all systems and operations. The human engineering program shall be conducted in accordance with the guidance of MIL-H-46855. Human engineering systems analysis shall include identification of those human engineering factors likely to require detailed study during the design engineering phase of the program. These analyses shall include such human performance requirements as those which exceed human capabilities, result in degradation of system objectives, and require performance of procedures which would result in proneness to human error. Human engineering design shall be accomplished for both the maintainer and the operator.

3.1.12 Weight distribution.

3.1.12.1 Four by two, structural truck. The weight on the front axle of a fully loaded truck, including the weight of four 275 pound firefighters, including gear and properly seated, shall be 40 percent, plus or minus five percent, of the weight of the truck.

3.1.12.2 Four by four, structural truck. The weight on the front axle of a fully loaded truck, including the weight of four 275 pound firefighters, including gear and properly seated, shall be 50 percent, within plus zero to minus five percent, of the weight of the truck.

3.1.12.3 Unbalanced weight. The truck weight shall not be unbalanced or improperly distributed. The difference in load between the wheels on any axle shall not exceed 2.5 percent of the load on the axle for any loading condition. The application of weights to the chassis for the purpose of meeting the weight distribution from side to side or fore to aft is not acceptable.

3.1.13 Body lean. The truck shall not lean to either side more than one percent of the width of the truck in any operational configuration. Body lean measurements shall be made at the outer most portion of the truck, excluding mirrors, mirror support brackets or lights, directly above each axle. The differences in height shall not exceed one percent of body width at that location.

3.1.14 Waterproofing. All mechanical and electrical components shall be arranged or protected so that their normal function will not be impaired by heavy rains, road splash, formation of condensation, pressure cleaning, spillage of extinguishing agents or leaks in the piping system.

3.1.15 Electromagnetic interference. The vehicle shall comply with the requirements of MIL-STD-461, Class C1, Group II for electromagnetic interference, except that the limits for UMO3 (Figure 8-1) shall be relaxed by 20 dB.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.1.16 Sound level.

3.1.16.1 Exterior sound level. Except for the siren and horn, the maximum sound level produced by the vehicle shall not exceed 84 dBA when measured in accordance with SAE J366. A 2 dBA tolerance shall be permitted for production units to provide for variation in test site, temperature, test equipment, and inherent differences in nominally identical vehicles.

3.1.16.2 Cab interior sound level. The maximum sound level, including siren and horn, in the interior of the cab shall not exceed 84 dBA when measured in accordance with SAE J336a.

3.1.17 Air transportability. The vehicle shall be air transportable in accordance with MIL-STD-1791 in a Type C-5 or C-17 Aircraft. The air transportable configuration shall include all auxiliary equipment, foam tanks full, and water in the agent system, except the water tank shall be empty. Removal of mounted equipment shall not be permitted to meet the required dimensions for air transport.

3.1.17.1 Tie downs. The truck shall have sufficient tie down points to be restrained in all directions during air transport. The tie down points shall be designed for use with attachments in the aircraft as described in MIL-STD-1791. There shall be only one tie down device for any aircraft tie down ring. Each end of each tie down device shall terminate at a separate attachment point without passing through any other attaching point. Sufficient clearance shall be maintained around all tie down points to allow for restraining chains without interference from any other structure. The tie down points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook. All tiedown points shall be rated at a minimum of 25,000 pounds and shall be marked for capacity. Tie down devices must comply with MIL-T-25959 for chains or MIL-T-27260 for straps.

3.1.18 Commercial components. Commercial components shall be used where possible and shall not be used outside their published ratings.

3.1.19 Altitude. Except as otherwise specified herein (see 6.2), performance requirements shall be based on operation at 3,000 feet above sea level.

3.1.20 Environmental conditions. The truck shall be capable of satisfactory performance under the following conditions without detrimental effect to subsequent operation:

a. Starting and operating in ambient temperatures ranging from -20°F to +125°F.

b. Starting and operating in an ambient temperature of -40°F when winterized as specified in 3.7.2.

3.1.21 Components. The truck shall consist of the following major components:

ITEM OR SYSTEM NAME	SEE REQUIREMENT
Chassis	3.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

Fire Pump	3.5
Body	3.6
Auxiliary fire fighting equipment	3.6.11
Stored water system	3.6.12

3.2 Performance.

3.2.1 Mobility. The fully loaded truck, including four 275 pound firefighters in full gear, or unless otherwise specified herein, shall be capable of the following:

a. The truck, in fully operational configuration, shall negotiate in both directions a 17° ramp connecting two horizontal surfaces. The ramp length shall be at least 1.2 times the wheelbase of the truck. The truck shall be capable of being driven forward, or backed from one horizontal surface across the ramp toe, ascend the ramp, cross the ramp crest and proceed on the other horizontal surface without any portion of the truck scraping any of the three surfaces.

b. Accelerate to a speed of not less than 65 MPH on dry, level, paved roads.

c. Accelerate from a standstill to a speed of 35 MPH in 20 seconds on dry, level, paved roads.

d. Ascend and descend 30 percent grades (4X2), 50 percent grades (4X4) both forward and reverse without adversely effecting vehicle operation or causing spillage of water, fuels, or lubricants.

e. Negotiate 20 percent side slopes, in both directions, without adversely affecting vehicle operation or causing spillage of water, fuel, or lubricants.

f. Be held and controlled by the service brakes in a heading up and heading down position on: 30 percent grade (4X2), 50 percent grade (4X4).

g. Be brought to a complete stop within 30 feet from a speed of 20 MPH on a dry, paved, approximately level surface free of loose material.

h. The parking brakes shall be capable of holding the truck motionless when headed in either direction on: 30 percent grade (4X2), 50 percent grade (4X4).

i. Traverse paved and unimproved roads, under any load conditions, as follows without adversely effecting vehicle components or installed equipment:

(1) Paved roads at an average speed of 60 MPH.

(2) Graded gravel roads at an average speed of 35 MPH

(3) Hilly cross country terrain at an average speed of 15 MPH minimum (4X4 truck only).

j. Be fully functional and operational within 15 seconds after starting engine when air brake system is properly pressurized.

k. Have a wall to wall turning diameter of 35 feet maximum in either

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

direction.

l. Have an angle or approach of at least 20° (4X2), 30° (4X4).

m. Have an angle of departure of at least 20° (4X2), 30° (4X4).

n. Have an interaxle clearance angle of at least 12° (4X2); 15° (4X4) with a minimum clearance at mid-wheelbase of 12 inches (4X2); 16 inches (4X4).

o. Maintain a speed of 55 MPH on a straight incline/grade of three percent.

3.2.2 Agent discharge. The truck shall be capable of discharging agent selectively from all fire pump discharge outlets. The truck shall be capable of:

3.2.2.1 Four by two, structural truck.

a. Discharging agent at a minimum rate of 375 GPM through any three of the 2 1/2 inch discharge outlets while pumping agent at a rate of 1,250 GPM at 150 psi from draft.

b. Discharging agent at a rate of 200 GPM through all of the 1 1/2 inch preconnected discharge outlets while selectively pumping from draft, hydrant, or the installed agent tank supply.

c. Discharging agent at a rate of 60 GPM through the booster hose while pumping from draft, hydrant, or the installed agent tank supply.

d. Interlocked to operate "stationary pumping" with the parking brakes set, the transmission in "neutral" position and the structural panel master switch activated.

3.2.2.2 Four by four, brush-structural truck.

a. Discharging agent at a minimum rate of 375 GPM through any three of the 2 1/2 inch discharge outlets while pumping agent at a rate of 1,250 GPM at 150 psi from draft.

b. Discharging agent at a rate of 200 GPM through all of the 1 1/2 inch preconnected discharge outlets while selectively pumping from draft, hydrant, or the installed agent tank supply.

c. Discharging agent at a rate of 60 GPM through the booster hose while pumping from draft, hydrant, or the installed agent tank supply.

d. Discharging agent through the booster hose at a rate of 60 GPM and any two 1 1/2 inch outlets at a rate of 200 GPM each at a minimum pump pressure of 150 PSIG during continuous forward motion and stop/start operation of the fully loaded truck from a level hard surface while not exceeding five MPH or while ascending a 50 percent grade at not less than one MPH.

e. Interlocked to operate "pump and roll" in the transmission's lowest forward speed and interlocked to operate "stationary pumping" with the parking

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:47Z
Check the source to verify that this is the currentversion before use.

brakes set, transmission in "neutral" position and the structural panel master switch activated.

3.3 Chassis. The (4X2) chassis shall be diesel engine powered and shall have a minimum GVW rating of 30,000 pounds. The chassis shall conform to the applicable requirements specified for a Type 1, Class C, truck described in KKK-T-2108 unless otherwise specified herein or in the contract documents. The (4X4) chassis shall be diesel engine powered and shall have a minimum GVW rating of 32,000 pounds. The chassis shall conform to the applicable requirement specified for a Type 1, Class D, truck described in KKK-T-2110 unless otherwise specified herein or in the contract documents. The total gross weight of the complete truck shall not exceed the manufacturer's GVW rating when fully equipped as specified herein, including all specified fire fighting equipment, serviced with fuel, lubricants, and agent tanks filled to capacity, 1,500 pounds in the hose bed and four 275 pound firefighters, including gear.

3.3.1 Engine. The truck shall be equipped with an EPA certified turbocharged diesel engine. The engine shall perform as specified herein when operated on diesel fuel fuel conforming to VV-F-800 and grade JP-8 turbine fuel conforming to MIL-T-83133. Calculations shall be furnished to verify engine selection in accordance with SAE J1349. The engine shall perform as specified herein, except for maximum speed, when operating on Grades JP-4 and JP-5 turbine fuel in accordance with MIL-T-5624. Any engine that is adversely affected by any of the above fuels is not acceptable. The engine shall be capable of providing the necessary power when using any of the above fuels, such that the normal progression of all transmission speeds shall occur. The engine shall not be adversely affected by long periods of idle or low speed operation. The engine shall be equipped with an emergency shut-down device. The engine shall be fully functional and operational within 15 seconds after starting.

3.3.1.1 Engine lubrication system. The engine lubrication system shall function satisfactorily on grades of up to 60 percent and side slopes of up to 20 percent. The oil filter shall be spin-on type.

3.3.1.2 Engine cooling system. The engine shall be equipped with a cooling system capable of satisfactory performance under the environmental conditions specified herein. The cooling system shall be of the pressurized type, of sufficient capacity to maintain a cooling liquid temperature of not less than 140°F and to provide proper cooling without overheating or loss of coolant when operated in an ambient temperature range of -20°F to +125°F during operating conditions of either stationary pumping or mobile performance as specified. The cooling system shall not require supplemental cooling (see 3.3.1.2.2 Heat exchanger) at ambient temperatures below 90°F.

3.3.1.2.1 Coolant system and indicators. The coolant system shall include a water filter, a deaeration system and a surge tank, or a coolant recovery reservoir of not less than two quart capacity. A high coolant temperature or low coolant level alarm buzzer and red indicator warning light shall be provided on the cab dash instrument panel and on the pump operator control panel. The coolant system shall be filled to capacity with a clean solution of ethylene glycol and water in equal parts by volume. A warning tag shall be securely attached to the radiator filler neck with the information "COOLING SYSTEM FILLED WITH WATER AND ETHYLENE GLYCOL IN EQUAL PARTS BY VOLUME - DO NOT DRAIN."

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.3.1.2.2 Heat exchanger. An auxiliary heat exchanger shall be provided that will permit delivery of water from the fire pump to the heat exchanger for cooling the engine liquid coolant without admixture. The heat exchanger discharge line shall be of adequate size to minimize back pressure. The flow of water from the fire pump to the heat exchanger shall be valve controlled. The valve control shall be located at the fire pump operator's control panel and properly identified.

3.3.1.2.3 Drain cocks. Drain cocks shall be installed at the low point of the cooling and heat exchanger systems and at any other points necessary to completely drain the systems. Drain cocks shall remain closed under the operating conditions encountered by the truck. Drain lines shall be furnished and shall discharge to the ground at a point below the bottom of the chassis frame.

3.3.1.3 Engine starting system. The engine shall start within 15 seconds in any ambient temperature within the required operating range of the vehicle. Installed glow plugs, fluid starting aids, and heat from the winterization system may be used to facilitate engine starting under the following conditions:

TEMPERATURE RANGE (Degree F)	STARTING AID PERMITTED
40 through 125	None
-20 through 39	Glow plugs and fluid starting aids
-40 through -21	Glow plugs, fluid starting aids, and heat from the winterization system.

A starting system interlock shall be provided to prevent starter engagement when the engine is running. An oil pressure switch shall not be used in the interlock. A starting system interlock shall be provided to prevent starting the engine when the transmission is in any position other than neutral.

3.3.1.4 Engine hourmeter. An hourmeter having a totalizing mechanism of not less than 9,999 hours shall be provided to accurately register the number of hours of engine operating time. The meter shall be of rugged construction to insure continuous trouble free performance under the most extreme equipment operating conditions specified herein. The meter shall be so mounted on the truck that it can be conveniently and easily read by personnel servicing the truck.

3.3.1.5 Engine brake. When specified (see 6.2), the engine shall be provided with an engine compression brake or an engine exhaust brake system. Proper operating instructions and warnings shall be located within easy view of the driver.

3.3.1.6 Exhaust system. Exhaust system shall conform to DOT Federal Motor Carrier Safety Regulations, Section 393.83. Muffler and exhaust pipe shall be insulated and shielded where radiated heat may damage electrical wiring or other truck mounted equipment. Muffler and tailpipe shall be aluminized steel and shall be assembled with anti-seize compound. The exhaust system discharge shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

be directed away from the fire pump operator's control panel area, away from any body panels and shall not be directed straight down to the ground.

3.3.1.6.1 (4X4 Truck Only). The entire exhaust system shall be protected from damage which could result from traversing rough cross-country terrain.

3.3.1.7 Fuel system. The fuel system shall conform to DOT Federal Motor Carrier Safety Regulations, Sections 393.65, 393.67, and as specified herein.

3.3.1.7.1 Fuel tank. The fuel tank shall be of not less than 50 gallon capacity. Tank and fill piping shall be so placed as to be protected from mechanical injury. The fuel tank filler shall be located for ease in filling, shall not extend outside the body skin, shall provide a drain to the ground for any spillage and shall be provided with spring loaded access door flush with the outer body skin and identified accordingly. The filler opening shall be large enough and arranged to accept an MD-3 aircraft fuel filler nozzle in accordance with MIL-N-87963.

3.3.1.7.2 Fuel filter. At least one filter of the spin-on fuel-water separator type shall be located in the fuel line between the tank and the engine. It shall be installed so that the element can be readily serviced without removing surrounding equipment. Unless the filter is mounted higher than the fuel tanks, a shutoff valve shall be included in the inlet line to the filter.

3.3.1.7.3 Fuel pump. A manually switched electric fuel priming pump shall be provided to assist engine restarting. The manual switch shall be located near the air bleed point on the fuel system.

3.3.2 Transmission. The truck shall be equipped with an automatic, electronic controlled transmission. The transmission shall be rated for heavy duty service and shall be designed to match engine torque and speed to the load demand. The transmission shall provide for driver selection of individual gears or ranges of gears required to meet performance requirements specified herein. The shift control shall incorporate a lighted indicator for each gear or range selected. The transmission shall be protected from an inadvertent operator shift into reverse. An electronic interlock shall be provided to prevent power transmission to vehicle drive during stationary pumping operations. All power takeoff connections shall be designed to produce and withstand at least 150 percent of the power and torque requirements.

3.3.2.1 Torque converter temperature gauge. (4X4 trucks only) A transmission torque converter oil temperature gauge shall be furnished. In addition to the gauge, a flashing red light and an audible warning device shall be provided. The warning light and audible device shall be activated when the converter oil temperature exceeds the transmission manufacturer's recommended safe operating temperatures. The gauge and warning light shall be suitably identified.

3.3.3 Transfer case. (4X4 trucks only) A system shall be provided to drive the front and rear axles. All wheel drive systems shall be designed to compensate for differential torque and speeds between the front and rear axles. An automatic or driver controlled differential locking device shall be incorporated into the differential. If required, an operator control, with indicator light on the instrument panel, shall be located in the cab.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.3.4 Axles. The load rating of each axle shall not be less than the imposed load on the axle, measured at the ground, with the vehicle at gross weight. Axle torque capacity shall be sufficient to spin the tires under all possible service conditions, including a 1.0 coefficient of friction. Axle design shall be such that, in the event of shaft failure, the wheel is not released from the vehicle. Maximum interchangeability of axle components shall be provided. Each axle shall be equipped with an automatic or driver controlled differential locking device. If required, an operator control, with indicator light on the instrument panel, shall be located in the cab.

3.3.5 Suspension. The truck shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the truck loaded to rated GVW. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Double action, heavy-duty shock absorbers shall be provided at front axle (4X2) and both axles (4X4).

3.3.6 Steering system. The truck shall be equipped with a power steering system, with manual steering available in the event of power assist failure. The force required for turning the wheels from lock to lock, with the vehicle at gross load on dry level concrete and the engine idling, shall not exceed 15 pounds when measured at the steering wheel rim. The steering system shall not require more than four turns lock to lock. A tilting and telescopic steering column shall be provided.

3.3.6.1 Steering wheel. The steering wheel shall have a maximum diameter of 18 inches. Not less than four inches of clearance shall be provided between the steering wheel rim and the nearest projection with the steering column in any position.

3.3.7 Wheels, Rims, Tires and Tubes. Rim and tire ratings shall conform to Tire and Rim Association recommendations for the type and size of the tires furnished. Tires shall be in accordance with MIL-M-8090. Individual tire loading at gross vehicle weight shall not exceed the Tire and Rim Association's load rating for highway operation. Tires shall be of commercial design and available to the general public. Sole source, special purpose design tires shall not be allowed.

3.3.7.1 (4X2) structural trucks. Tires shall be steel radial design with nondirectional extra traction tread design. Tube type tires may be used. A spare wheel and tire assembly shall be furnished, but a spare carrier is not required.

3.3.7.2 (4X4) structural truck. Single wheels shall be provided on both the front and rear axles of this type truck. Tires shall be tubeless steel radial design with nondirectional off road traction tread design capable of sustained highway speeds. A spare wheel and tire assembly shall be furnished, but a spare carrier is not required.

3.3.7.3 Inner tubes. For tube tires, inner tubes shall be of heavy duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with "TRA" recommendations.

3.3.8 Brakes. Brakes shall conform to DOT Federal Motor Carrier Safety

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

Regulations, Sections 393.40 through 393.42, 393.43, 393.45 through 393.52 and as specified herein.

3.3.8.1 Service brakes. The service brakes shall be four wheel, automatic adjusting, air application type. Brake fade must be a design consideration for the operating conditions and the truck's intended use.

3.3.8.2 Brake system air compressor. The capacity of the air compressor shall be sufficient for the intended service but shall be not less than 12 cubic feet per minute. The air compressor shall be equipped with a governor and the compressor intake shall be properly connected to an air cleaner. An air dryer shall be furnished between the compressor and the quick build-up tank. The moisture removed from the air by the air drying system shall be disposed of automatically. Moisture shall not bypass the air dryer during moisture ejection.

3.3.8.3 Brake system air storage reservoirs. Adequate air storage reservoirs with a buzzer type low pressure indicator and a safety relief valve shall be provided. The system shall be of the quick build-up design with the air compressor connected to a small reservoir which is pressure relieved into the larger reservoirs when the minimum pressure is reached for optimum operation of the brake system. The time required to reach build-up pressure from zero gauge pressure and at the engine idle speed shall not exceed 30 seconds. Time controlled high idle (maximum 30 seconds) may be provided to meet the minimum brake air operating pressure. If other than brake air supply requirements are supplied from the reservoirs, a priority system shall be installed to assure that an adequate supply of air is always available to operate the brake system. The quick build-up reservoir shall be equipped with an automatic drain valve and all other reservoirs shall have manual drain valves. An auxiliary air line, with an automatic ejector type disconnect coupling at the right hand rear corner of the vehicle, shall be provided for charging the air tanks. A mating 1/2 inch air line connector shall be provided loose with each truck.

3.3.8.4 Parking brakes. Parking brakes that are independent from the air service brake system shall be provided. The operating controls of the parking brakes shall be independent of the operating controls of the service brakes. The parking brake shall be capable of meeting the performance requirements specified.

3.3.8.5 Air brake emergency system. An emergency brake system shall be furnished. The system shall be capable of being applied from the driver's seated position. A manual or mechanical means to release the brakes shall be provided when compressed air is not available to release the brakes. The emergency system shall also serve as a parking brake and shall conform to DOT Federal Motor Carrier Safety Regulations, Sections 393.41 and 393.52.

3.3.8.6 Air Horn. Dual single trumpet air horns, with individual controls located between the driver and crew chief, shall be provided. The air horns shall be located behind and project through the front bumper, one near each end.

3.3.9 Electrical system. The vehicle shall be equipped with a 12 volt (V), negative ground, electrical system. Wiring shall be waterproof and arranged so that their normal functioning will not be impaired by entrance of water during heavy rain, road splash, condensation or pressure cleaning. Wiring shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

color coded or numbered for identification and consistent throughout the truck. All circuits shall be protected by automatic reset circuit breakers located in the cab and accessible for service. The contractor shall certify that corrosion prevention is in conformance with MIL-STD-1250, except when in conflict with MIL-STD-889 with respect to dissimilar metals, where MIL-STD-889 shall prevail. All electrical components shall conform to MIL-STD-454, requirement 1. Printed circuit/printed wire boards shall be conformal coated with materials in accordance with MIL-I-46058. A battery isolation switch shall be provided to disconnect power to the main buss and to provide isolation between battery banks. A battery pilot indicator light shall be provided for each battery bank, within view of the driver's seated position, that illuminates to indicate a closed contact to the main buss.

3.3.9.1 Alternator. The truck alternator shall be furnished complete with a rectifier and voltage regulation system capable of supplying a minimum of 200 amperes DC power. The system shall be capable of supplying the full power requirements for operation of the truck, lights, warning lights, radio circuit, winterization system and all equipment and accessories at engine idle. Provide a reverse current relay to permit operation of the winterization system when the truck engine is shut off and the battery charge is being maintained from an external, 110V AC, power source.

3.3.9.2 Battery system. A dual 12 volt battery system shall be furnished. Each battery in the system shall be 12 volt maintenance free with characteristics listed in W-B-131. Each battery shall be rated in terms of cold cranking capacity at $0^{\circ}F$ in accordance with SAE J537. Each battery shall have not less than the engine manufacturer's recommendation of cold cranking amperage required at $-20^{\circ}F$ for the engine furnished. Battery compartments shall be located as close as possible to the engine in order to reduce battery cable length. The battery cables shall be protected by use of grommets where the cables pass through metal bulkheads and shall be covered by a protective boot where the cables are attached to electrical switches or connectors. The battery compartments shall provide full access to the batteries for service and shall be ventilated and treated against acid fumes and spillage. The batteries shall be mounted in acid resistant trays. A sliding carriage shall be provided if the batteries must be moved sideways to permit access for servicing or replacement.

3.3.9.3 Battery charger and battery charging receptacle. A 110/220V AC powered, 20 amp or larger output, battery charger shall be provided and mounted on the vehicle to provide charging of the truck batteries. A recessed, automatic ejector type, external power receptacle with cover shall be installed on the driver's side at the rear of the truck body and wired to the battery charger. A 25 foot electrical cable, with connector to match truck receptacle on one end and three prong 110V AC connector on the other end, shall be provided with the truck. A 110/220V European power adapter shall be provided loose with each truck. The battery conditioner shall operate with the battery master switch (see 3.3.9) in either open or closed position. A pilot light, located above the power receptacle, shall indicate when the charger is connected to the 110/220V AC power source.

3.3.9.4 Lighting system. The lighting system shall be such that one circuit will not be affected by a short in any of the other circuits. All metal rims, shells, mounting arms, and bases of all exterior lights shall be chromium plated, except when painted Forest Green (see 3.10.1) or Desert Sand (see

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.10.2). The stoplights and taillights shall be recess mounted with the lenses flush with the body. All lights shall be so mounted as not to obscure mounted equipment, and to prevent damage due to operation, removal or replacement of firefighting equipment. Clearance and side marker lights shall be provided and shall be combined. The following additional lighting shall be furnished.

3.3.9.4.1 Spotlights. Two automotive type sealed beam spotlights shall be furnished, one on each side near each front corner on the cab roof. Controls shall be extended inside the cab for on-off control and movement of the lamp housing through a horizontal included angle of at least 270°, and a vertical angle of at least 90° above and 35° below horizontal. The spotlight control handle shall not obscure the driver's vision, or interfere with normal driving functions or entering and exiting the cab.

3.3.9.4.2 Hose pickup lights. Two adjustable sealed beam hose pickup lights shall be mounted on the rear handrail supports. These lights shall be activated by a single weatherproof switch not more than 60 inches from the ground.

3.3.9.4.3 Floodlights. Two each 12 volt telescopic quartz floodlights shall be installed at the back of the cab and accessible from the walkway. Telescopic tubing shall be stainless steel or anodized aluminum. The floodlights shall be readily adjustable to a minimum of four feet above the cab roof and shall lock securely in place. The lights shall be individually fused and powered through the truck electrical system.

3.3.9.4.4 Light bar. A light bar, Federal AeroTwinSonic Model 24XQ, or equal, shall be securely mounted on the roof of the cab. The light bar shall have all red lights and shall be controlled by a switch located on the vehicle instrument panel. When specified (see 6.2) the light bar shall have all blue colored lights.

3.3.9.4.5 Engine compartment lights. Two shielded lights, arranged to illuminate both sides of the engine, and a switch shall be located in the engine compartment.

3.3.9.4.6 Back up lights. Two automatic backup lights, not less than five inches in diameter, recess mounted at the rear of the body, shall be provided.

3.3.9.4.7 Pump operator's control panel lights. (see 3.6.3.).

3.3.9.4.8 Warning lights. Two red alternating flashing 360° strobe warning lights shall be mounted on the rear handrail supports adjacent to the hose pickup lights. The lights shall have a combined minimum flash rate of 120 flashes per minute. The lights shall be controlled by the Light Bar switch (see 3.3.9.4.4). When specified (see 6.2) the warning lights shall be blue colored.

3.3.9.4.9 Equipment compartment lights. (see 3.6.9)

3.3.9.4.10 Forward flashing lights. Two forward flashing red strobe lights shall be mounted on the front face of the truck cab below the windshield. The lights shall flash alternately with a sequence of 60 cycles per minute. The lights shall be controlled by the Light Bar switch (see 3.3.9.4.4). When specified (see 6.2) the flashing lights shall be blue colored.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:47Z
Check the source to verify that this is the current version before use.

3.3.9.4.11 Alternating headlight switch. A separate switch, accessible to the driver, that causes the truck headlight high beams to flash alternately at 60 cycles per minute shall be provided. Low beam headlights shall not be affected by the alternating lights.

3.3.9.4.12 Service lighting. In addition to the step and walkway lighting requirements of NFPA 1901, lights shall be provided for illuminating the rear platform, side platforms and all suction and discharge connections around the truck. All service lighting shall be recessed or otherwise protected from damage during firefighting operations. All service lighting shall be switched from both the cab instrument panel and the pump operator's control panel.

3.3.9.5 Backup warning device. An automatically activated electric/electronic pulsating signal warning device shall operate while truck is in reverse gear. The backup alarm shall be in accordance with SAE J994.

3.3.9.6 Siren loudspeaker. A commercial 12V combination electronic siren and loudspeaker, Federal PA-300, or equal, shall be provided. The system shall have a minimum 100 watt output and shall be compatible with all current US military installed radio systems using either carbon or magnetic microphones. The system shall include a horn/siren switch and shall have controls for radio, manual, wail, yelp and hi-lo operation. The operating switches and controls shall be accessible to the driver and crew chief in their normal seated positions. A Federal Model CP100S speaker, or equal shall be mounted on the top of the front bumper. The speaker shall be protected by a brush guard on 4X4 trucks. Dual hand and foot operated switches shall be provided for both the driver and crew chief.

3.3.9.7 Communication equipment. A dedicated electrical circuit with 15 amp breaker and wire shall be provided to the designated location for base radio (see 3.4.1). In addition to the radio communication system:

3.3.9.7.1 Provide a compact weatherproof speaker with adjustable volume control installed at the pump operator's control panel.

3.3.9.7.2 Provide a weatherproof microphone at the pump operator's control panel.

3.3.9.7.3 Provide a headset jack at the pump operator's control panel. The headset shall interface with the radio and shall be capable of transmitting and receiving functions.

3.4 Cab. A weathertight enclosed, four passenger, four door, tilt or cab forward, cab conforming to the recommendations of NFPA 1500 shall be provided. A cab forward design shall include an adequate gasketed opening in the roof, with a removable weathertight cover, to allow removal and replacement of the engine/transmission assembly when required. A removable cab using quick disconnect fittings for mechanical systems and waterproof cannon type connectors for electrical lines is considered equal to the removable roof cab design. A tilt cab shall have a hydraulically powered assist mechanism to raise or lower the cab within 30 seconds. The cab shall provide an arrangement of unobstructed seating and headroom for four fully clothed firefighters. The cab shall be sound and heat insulated from the engine and the elements. Vapor and gas proof

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

access panels or doors shall be provided in the engine cover to inspect and fill all service fluids without removal of or tilting the cab. The cab and truck body shall be the same width.

3.4.1 Cab construction. The cab shall be of all aluminum, reinforced, welded construction. Aluminum sheets used in construction of the cab shall be a minimum of 0.125 inch thickness. The cab roof shall be fabricated of diamond tread plate. The cab floor and steps shall be fabricated of diamond tread plate. Steps shall be provided at each cab door and shall be within view and access of personnel entering and leaving the cab. The first step shall be not more than 22 inches above the ground. Each step shall be of maximum width and depth to safely accommodate arctic type or size twelve firefighting boots. The steps shall support the weight of a 275 pound firefighter, including gear, without permanent deformation. The cab steps, if exposed to the elements, shall be fabricated of a drainable open tread. Rubber boots shall be provided for all control openings in the floorboards. Full circular wheel well liners shall be required. Mudflaps shall be provided on the rear of each front wheel well to prevent dirt and water entry onto cab steps and walkway. Grab handles or bars shall be provided on each side of the cab, at each door, to assist entry into the cab. A grab handle shall be contoured to match the rear facing edge of the cab from the walkway step, over the top to the opposite step and attached with four inch stand off brackets. A space at least 18 inches long by 12 inches wide by 12 inches deep shall be provided for mounting a radio for base operations between the driver and crew chief and shall not be in the footpath area. The cab shall comply with the visibility requirements of SAE J1050a. The cab shall be equipped with two sun visors. Cabs fabricated from all stainless steel materials, including skin panels, inner panels, reinforcements, and tread surfaces shall be considered equal, however, the cab and truck body shall be fabricated of the same materials.

3.4.2 Cab doors. Cab doors shall be provided for all seated positions. Cab doors shall be constructed of the same material as the cab and shall be equipped with roll down or sliding windows. A fixed window shall be provided in the right side front door that gives the driver a view of a private automobile positioned to the right of the cab. Cab door hinges shall be fabricated of unpainted stainless steel and shall be attached to the door and post with bolts and nuts. Each door shall provide entrance and exit of personnel with physical dimensions in the range of small to large, as defined in SAE J833, while wearing firefighting gear, including self contained breathing apparatus. The outside door handles shall be accessible to personnel standing on the ground and shall be designed for operation by personnel wearing arctic mittens as specified in SAE J925. The latch shall be designed to prevent entry of water and ice into the mechanism. Door locks shall not be provided. Water tightness of the cab, doors and windows shall be checked by Method 506.2, Procedure I of MIL-STD-810.

3.4.3 Windows. All windows in the cab shall be in accordance with FMVSS 205. All cab glass shall be tinted and sized to provide maximum visibility from the cab and cab doors. All sliding window channels shall be lined to prevent rattles.

3.4.4 Windshield wiping and washing systems. The truck shall be equipped with an electric operated windshield wiping and washing system in accordance with FMVSS 104. The wiping system, in conjunction with the defrosting system, shall provide clear vision for the cab at the temperature extremes specified. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

wiper arms and blades shall be of sufficient length to clear not less than the windshield area described by SAE J198. Wiper controls shall return the wiper blades to a park position, out of the line of vision, when not in use.

3.4.4.1 Windshield washers. A windshield washer system shall be furnished. The fluid container, with a minimum of one gallon capacity shall be constructed of a translucent material in order that the fluid level may be observed without removal of the cap. The washer control shall be incorporated into the wiper switch. The washer nozzles shall be incorporated into the windshield wiper arm assembly (wet arm).

3.4.5 Climate control system. The climate control system shall provide heater, defroster and air conditioner modes. The system shall include a heater core(s), a blower(s), a refrigerant compressor, a condenser, an evaporator(s) and the plumbing, ductwork, dampers, controls and other components necessary to meet the performance requirements specified herein. The climate control system shall indraft at least 60 cubic feet per minute of filtered fresh air into the cab. A manual emergency damper shall shut off flow of outside air when operating in a smoky environment. At least four adjustable panel louvers shall be provided, approximately evenly spaced across the width of the cab; each shall provide approximately equally air flow. Cab mounted components shall be protected from inadvertant damage by personnel footgear.

3.4.5.1 Heater and defroster. The heater shall maintain a temperature of at least 40 degrees F at the floor with an ambient temperature of -40° F. When occupied by four crew members, the defroster shall keep at least 75 percent of the windshield and side windows clear with an ambient temperature of -40° F. The defroster shall be in accordance with FMVSS 103."

3.4.5.2 Air conditioner. The air conditioner shall, at normal engine idle speed (standby condition), be capable of reducing the cab interior temperature to 95° F within 30 minutes with the outside ambient temperature at 125° F. The vaporator condensate shall drain to the ground under any operating condition.

3.4.6 Mirrors. Outside rearview mirrors shall be mounted on each side of the cab. The mirrors shall be a low profile mirror and the top of the mirror shall not impair the horizontal line of vision of a fifth percentile woman sitting in the driver's seat. Combination (flat and convex) mirrors, enclosed in a common housing, shall be furnished. Each combination mirror shall have at least 60 square inches of flat reflective area and a convex surface having at least 30 square inches of reflective area. Each mirror shall have at least three supporting arms or a heavy duty spring loaded ratchet type mounting bracket for side mounting. When specified (see 6.2) outside rearview mirrors shall be heated. Convex mirrors, four inch minimum diameter, shall be mounted on the front face of the cab to gives the driver and crew chief full view of the ground immediately in front of the front bumper.

3.4.7 Interior lighting. Door operated dome lights shall be provided in the the cab above each seated location. Step lights into the cab shall also be activated with the door switch. The dome lights/step lights shall have a master switch to prevent automatic door activation. An additional adjustable aircraft type reading light, with locally accessible switch, shall be directed toward each seated position in the cab.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.4.8 Seats. All seats shall be upholstered over deep air foam cushions with first grade black vinyl covering suitable for fire service. Seat belts, in accordance with FMVSS 207, with non-cinching automatic retractors shall be provided and installed in accordance with FMVSS 209 and 210 for each seated position. Seat belts shall be of sufficient length to accommodate firefighters wearing full gear. The seated area shall provide unobstructed leg, knee and foot room for the 95th percentile man dressed in firefighting gear. The driver's seat shall be a heavy duty, four way, adjustable air ride bucket type. All other firefighter's seats shall be Superior First In or equal with padded flanges on each side and forward of the self contained breathing apparatus. The firefighter's seats lower cushions shall be not less than 18 inches wide and shall provide room for a firefighter wearing full gear and the self contained breathing apparatus. The backs of each firefighter's seat shall have a walk away bracket, Ziamatic Model UN-6-30-3-F, or equal, for mounting of a self contained breathing apparatus with regulator in either inverted or upright positions. A series of predrilled holes shall be provided in the firefighter's seat backs for vertical adjustment of the walk away brackets. Each firefighter's seat shall have a padded headrest and a storage tray for the self contained breathing apparatus face mask. The self contained breathing apparatus shall provide 60 minutes of service and conform to all NFPA requirements.

3.4.9 Coat Hook. Four hooks of sufficient strength to hold a coat and helmet shall be provided in the cab, accessible from each seated position.

3.4.10 Controls and operating mechanisms. All cab controls and operating mechanisms shall be arranged for left hand drive. Controls shall be within reach of the driver's seated position. Controls shall not obstruct entrance or exit from any seated position. Instruments and controls shall be identified as to function and installed to facilitate servicing. Instruments, controls and wiring thereto shall be incorporated into modular design panels and shall be removable by quick disconnects for ease of maintenance.

3.4.11 Instrument panel. The cab instrument panel shall be complete, with not less than the following minimum instrumentation and controls identified:

a. Indirect panel lights (to illuminate all cab switches, instruments, and controls.

b. Keyless starting switch with pilot light and safety switches or devices to prevent restarting a running engine or starting the engine unless transmission is in neutral.

c. Service lighting switch.

d. Compartment "door open" indicator light.

e. Windshield wiper and washer control switch.

f. Heater and defroster controls.

g. Master warning light control switch.

h. Headlight alternating flash switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

i. Speedometer with recording odometer.

j. Engine tachometer.

k. Engine oil pressure gauge with low pressure warning light.

l. Engine coolant temperature gauge with high temperature warning light.

m. Transmission converter oil temperature gauge with high temperature warning light.

n. Voltmeter with high and low voltage warning light.

o. Fuel level gauge.

p. Service brake and emergency brake air pressure gauge with low pressure audible alarm.

q. Parking brake and emergency brake switches.

r. Headlight high beam indicator light.

s. Signal buzzer. Buzzers shall be installed below instrument panel to receive a signal from the rear platform (see 3.6.8).

t. Master electrical disconnect switch with circuit indicator lights (see 3.3.9) shall be located in the cab within view and easy access of the driver.

u. Water tank level gauge with low level indicator light (4X4 only).

v. Foam tank level gauge with low level indicator light (4X4 only).

w. Indicator light for safe pumping modes for pump and roll operation (4X4 only).

x. Pump engage switch for pump and roll operation (4X4 only).

y. Agent pressure gauge (4X4 only).

z. Transfer case low range indicator light, if required, (4X4 only).

aa. Direct reading incline meters with safe slope warning light (4X4 only).

3.4.12 Front bumper and towing devices. A heavy duty full width front bumper and tow eyes shall be provided at the front of the truck. The front bumper shall be a polished or plated metal fabrication. The front bumper shall project out a minimum of 18 inches in front of the cab. Drainable troughs with hinged covers and latches shall be provide between the front of the cab and the front bumper. One of the troughs shall be for storage of the soft suction hose specified in 3.6.11.b. The space between the front bumper and cab shall be decked with aluminum diamond tread plate. The two front tow eyes shall be mounted to the frame or rigid members which are attached to the frame. Access to the two front tow eyes shall not be through the cab front panel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.4.13 Front mounted winch. (4X4 trucks only) A front mounted, driver controlled, hydraulic driven winch conforming to SAE J706, shall be mounted on the truck between the cab, the front bumper and below the decking. The winch shall be powered by an independent engine driven pump. The winch shall be a single drum type with one forward and one reverse speed. All winch controls shall be mounted inside the cab. The winch cable shall spool from the bottom of the drum and below the front bumper and shall not encroach on the specified approach angle of the truck. The winch shall have a rated single line pull capacity of not less than 20,000 pounds with a line speed between 15 and 30 feet per minute on the bare drum. The winch shall be wound with not less than 185 feet of 5/8 inch diameter preformed 6 X 37 improved plow steel wire rope with welded hook and shall also include a six foot long end chain with hook. The winch shall be equipped with a four way roller guide and an integral, adjustable safety brake. A winch drum guard shall be provided to confine the cable between the drum flanges. The guard shall consist of not less than 1/4 inch vertical side plates, conforming to the outside radius of the drum flanges with six bars, 3/8 by 1 1/4 inches welded to the side plates, three bars equally spaced on the top and bottom of the drum. Clearance between the winch drum and the guard shall be not more than one half the cable diameter.

3.5 Fire pump. A UL certified, single stage, centrifugal fire pump, conforming to NFPA Standard Number 1901, shall be midship mounted. The method used for driving the pump shall produce a minimum 150 percent of the maximum power required. The linkages for engaging the pump shall be guided and supported to minimize vibration, flexing, etc. Controls for engaging the pump drive for "pump and roll" shall be located in the cab (4X4 only). Threaded connections of the pump and pump accessories shall conform to Handbook H28. The pump shall be accessible for service and removal without dismantling of other components of the truck. When drafting at a minimum lift of 10 feet through 20 feet of 6 inch hard suction hose, the pump shall deliver agent capacities with corresponding discharge pressures specified below at less than the engine manufacturer's recommended governed speed of the truck engine:

Minimum Capacity (GPM)	Maximum Net Pump Pressure (PSI)
1,250	150
875	200
625	250

3.5.1 High temperature pumping. The pump drive, gearing and housing shall be such that, at rated capacity and pressure, the temperature rise in the gear lubricant will not exceed the maximum temperature recommended by the pump manufacturer for continuous operation in an ambient temperature of +125°F. A high temperature indicator light and warning buzzer shall be provided on the pump operator's control panel.

3.5.2 Pressure rating. The pump, suction and discharge headers and all parts subject to pressure, including the booster reel and hose, shall withstand a hydrostatic pressure of 350 PSI for a period of 10 minutes.

3.5.3 Pump drain system. A pump drain system shall be provided to drain the pump and all water carrying lines. Drain control shall be located on the pump operator's control panel. Drains shall be located inside the pump compartment

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

and shall discharge to the ground from a point below the bottom of the truck frame.

3.5.4 Primer pump. An electric motor driven rotary primer pump shall be provided. The priming pump shall be operable with a control on the pump control panel and shall have the capacity to prime a dry fire pump through 20 feet of six inch hard suction hose at a 10 foot lift and discharge water in not more than 30 seconds under standard conditions. With the suction connections capped, the primer pump shall be capable of developing a vacuum of 22 inches Hg at altitudes up to 1,000 feet and shall hold the vacuum with a drop not exceeding 10 inches Hg in 10 minutes with the pump shutoff. The primer pump shall be accessible for service and removal without dismantling of other components of the truck.

3.5.5 Suction and discharge connection. The pump shall be provided with the following suction and discharge connections. Connections shall be mounted with sufficient clearance to allow the use of spanner type wrenches by personnel wearing arctic type mittens. All suction and discharge connections shall have chained-on pin or rocker lug protective caps or plugs and shall be chromium plated brass or pyrolite. All suction and discharge valve controls shall be on the pump operator's control panel.

a. Three six inch male NSFHT suctions with removable strainers, removable Hydra-Shield Pre-Con, or equal, automatic inlet valves with bleeders, capable of delivering a minimum of 1000 GPM at 200 PSI net pump pressure, and chained on long handle caps. The suctions shall be located, one on each side of vehicle and one located at the curb side front of the truck cab. Each suction shall be located not more than 48 inches above the ground. The front suction may be routed through the front bumper.

b. Three 2 1/2 inch female NSFHT suctions with ball valves, removable strainers and chained on plugs, one on each side of vehicle adjacent to the six inch suctions and one at the curb side rear of the truck body. The suctions shall be located not more than 48 inches above the ground, except the rear suction may be measured from the platform. The suction connections shall have removable, swivel type, 45 degree elbows attached.

c. One valved connection between the pump suction manifold and the agent tank capable of delivering not less than 625 GPM from the agent tank.

d. Six discharges with ball valves and 2 1/2 inch male NSFHT connections and chained on caps, two on each side of the truck and two below the rear hosebed, one on each side. Discharge outlet centers shall be not more than 60 inches above ground level, except the rear discharges shall be not more than 48 inches above the platform. The discharge connections shall have removable swivel type 45 degree elbows attached.

e. One discharge with ball valve and 1 1/2 inch male NSFHT for the booster hose reel. Valve control shall be at the pump operator's control panel.

f. Three discharges with ball valves and 1 1/2 inch male NSFHT swivel connections for the transverse preconnected hoses.

g. One valved outlet large enough to flow 1,500 GPM to the monitor turret.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

h. One valved connection between the pump discharge header and the water tank large enough to allow a tank fill rate of 500 GPM.

3.5.6 Monitor. A 1500 GPM monitor (deluge gun) with handwheel controls shall be mounted in an area at the top of the pump operator's control panel. The monitor shall be an Akron, Elkhart or equal. A gasketed mating flange shall be furnished at the base of the monitor to permit easily removal. A cradle with a quick release fastening device shall be provided to hold the nozzle ends of the monitor when stowed. A separate stream shaper pipe of not less than 15 inches in length for use with stacked tips shall be furnished. Stacking type tips, 2 inch, 1 3/4 inch, 1 1/2 inch and 1 1/4 inch shall be furnished. The stream shaper and stacked tips shall be pyrolite or equal. The unplated components of the monitor shall be painted the same color as the truck. The handwheel controls shall be within reach of the pump operator when standing at the control panel and shall not interfere with all other controls.

3.5.7 Relief valve. An automatic relief valve shall be provided for controlling the pressure from all discharge outlets. The relief valve shall be accessible for maintenance without removal of other components. The relief valve control shall be located at the pump operator's control panel so that adjustments can be made by the pump operator from his normal position without the use of tools and in full view of the discharge pressure gauge. The relief valve shall be of such design that, when set 10 PSI higher than the pump working pressure, there will not be an increase in pump pressure greater than 30 PSI when discharge valves are closed slowly. The relief valve shall perform satisfactorily when pumping at full rated capacities. The relief valve drain line shall discharge to the ground at a point below the bottom of the chassis frame.

3.6 Body. The fire truck body shall be fabricated of all the same materials and shall include a walkway, a fire pump operator's control panel, three transverse hose beds, removable agent tanks, main hose bed, equipment compartments, rear platform, and firefighting equipment. The body shall be entirely separate from the cab. The body skin and structural members shall be aluminum alloy having not less than 42,000 PSI ultimate strength and 35,000 PSI yield strength with thickness of not less than 0.125 inch, or the body skin, structural members, and tread surfaces may be fabricated from stainless steel. The body shall be reinforced at all stress concentration points and where equipment and compartments are attached. The body shall be designed and constructed so it shall not entrap water or experience any penetrating corrosion for twelve years under the operating conditions specified herein. The body shall be free from rough and sharp edges. A continuous diamond tread plate drip rail shall be provided above all compartments. A diamond tread plate rub rail shall extend from the side to rear platform, except at the wheel openings, and project a minimum of 1 1/2 inches outside the side panels of the body, then under the truck body to eliminate a sharp vertical edge. Diamond tread plate shall cover the areas 12 inches up on the body above the side platforms and the entire area between the rear platform and hose bed, except for any compartment doors.

3.6.1 Walkway. A walkway shall be provided between the back of the cab and the fire truck body. The walkway shall be the full width of the pump operator's control panel and wide enough to accommodate the fire pump operator in full

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

firefighting gear. Access steps shall be provided on both sides of the truck below the walkway. Step areas shall provide ample space for firefighters wearing arctic or heavy boots. The walkway and steps thereto shall be fabricated of diamond tread plate and of sufficient strength and rigidity to sustain the shock loads imposed by 275 pound firefighters, including full gear, while jumping and running in this area without permanent deformation. Grab rails shall be provided to assist personnel in climbing or stepping onto the walkway area. A padded fold-away barrier shall be provided between the rear corner of the cab and the front edge of the truck body to restrain a firefighter from falling from the walkway. The forward end of the truck body, at the walkway, shall be covered with diamond tread plate. Equipment compartments, if possible, shall be configured into each side of the walkway and step enclosures. The compartment covers shall be diamond tread plate and hinged to open upward and shall be latched down. These compartments may be used for battery enclosures.

3.6.2 Fire pump compartment. The fire pump shall be located for ease of service and shall be removable intact through weathertight opening(s) without disconnecting more than the pump suction and discharge fittings. The backside of the pump operator's control panel shall be protected from the pump assembly. No other equipment, except removable firefighting hardware shall interfere with access to the fire pump. If the weathertight openings are located in a horizontal surface they shall be fabricated of diamond tread plate and reinforced to support a 275 pound firefighter, including full gear, while running or jumping in this area, without permanent deformation.

3.6.3 Fire pump control panel. A pump control panel shall be top center mounted and elevated to provide 360° visibility around the truck. Pump pressure and suction gauges connected to the pump manifolds shall be not less than 3 1/2 inch diameter. Pressure gauges shall be provided to the hose line side of each discharge valve. Gauges shall be of the single Bourdon type, liquid filled and flush mounted. All valve operating handles and other controls shall be firmly supported, realignable, noninterfering and operable by personnel wearing arctic type mittens. All manually operated valves shall be activated by rigid stainless steel rods or heavy wall tubes. Collars, guides and grommets shall be installed at all control openings in the panel, except push-pull type controls shall be provided with rubber, nylon or other type of corrosion resistant material to prevent rattles. All exposed metal parts, such as valve controls, gauge rings, and pressure controls, shall be stainless steel, polished aluminum, or chromium plated materials. Hand grip knobs on control handles may be black, temperature resisting plastic. The full width of the panel shall be illuminated with nonglare lighting and shall include the following:

a. Pump suction gauge - 30 inches Hg vacuum to 600 PSI.

b. Pump pressure gauge - 0 to 600 PSI.

c. Ten discharge outlet gauges located adjacent to the respective discharge outlet control handles.

d. Two plugged 1/4 inch NPT connections for pressure test gauges, one each on the pressure and the suction sides of pump.

e. Pump pressure relief valve control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

f. Primer pump control.

g. Master pump drain control.

h. Heat exchanger control.

i. Engine tachometer.

j. Engine oil pressure gauge and low oil pressure warning light and buzzer.

k. Engine coolant temperature gauge and high coolant temperature warning light and buzzer.

l. Engine throttle, pull type with latch and micro adjustments.

m. Water pump to tank fill valve control.

n. Water tank discharge and drain controls.

o. Water tank liquid level gauge with low level indicator light.

p. Water tank to pump valve control.

q. Pump intake valve controls.

r. Pump engage switch with indicator light and guard.

s. Structural panel activation switch and indicator light and guard.

t. Telescoping light switches.

u. Weatherproof speaker, weatherproof microphone and headset jack.

v. Foam tank level gauge with low level indicator light.

w. Foam system activation switch with indicator light.

x. Discharge pressure gauge for the monitor (deluge gun).

y. Fire pump high temperature warning light and buzzer (see 3.5.1).

z. An instruction plate of nonferrous metal or stainless steel, screwed or riveted to the pump operator's control panel, shall indicate the fire pump delivery rate at 150, 200 and 250 PSIG at sea level with the respective engine RPM's. All pumping controls, valve controls, instruments, switches and warning lights shall be identified with instruction and identification plates.

3.6.4 Hose bed. The hose beds shall have a capacity for stowing a minimum of 2,000 feet of three inch double jacketed (polyester) rubber lined fire hose or 1,000 feet of five inch double jacketed (polyester) rubber lined fire hose, whichever is greater, plus 1200 feet of 2 1/2 inch double jacketed (polyester) rubber lined fire hose and 600 feet of 1 3/4 inch double jacketed (polyester) rubber lined fire hose in horizontal layers. The 1 3/4 inch hose beds shall

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the currentversion before use.

have three transverse compartments capable of stowing a minimum of 200 feet of preconnected hose each. Three adjustable and reinforced hose bed dividers shall be provided. The divider partitions shall extend to the same height as the body side panels, but not higher than 72 inches above the side or rear platforms, and shall be removable. The hose beds shall have reinforced metal floors, not more than 54 inches above the side or rear platforms, designed to drain completely with the truck on a level surface. Drains, if required, shall extend below the truck frame. The hose bed floors shall support 200 pounds per square foot of area without permanent deformation. The 1 3/4 inch hoses shall be preconnected to pump discharge lines extending through the center floor of each transverse hose bed. The transverse hose connectors shall be swivel type, watertight and move a minimum of 45 degrees either side of vertical throughout the operating pressure range. Vertical roller guides shall be provided at the ends of the hose beds and on the ends of the hose bed dividers. A horizontal fixed handrail shall be located at the ends of and below the hose bed floors. Dri-Dek, or equal, shall be installed on the floors of all hose beds. Hose beds fabricated of elevated tubular aluminum slats, with no sharp edges or corners, above a solid metal floor and capable of supporting 200 pounds per square foot without permanent deformation shall be considered equal. Only one style of hose bed floor shall be allowed on the truck.

3.6.5 Side and rear platforms. The truck shall have side platforms extending from the walkway steps to not less than the rear edge of the transverse hose beds and not less than 18 inches deep. The rear platform shall extend across the full width of the truck body. All platforms shall be the same height, fabricated of diamond tread plate, not more than 22 inches above ground and level when the truck is fully loaded. The platform edges shall be flanged down a minimum of two inches and flanged under a minimum of one inch. The corners of the rear platform shall be rounded or chamfered, welded and ground free of sharp edges. The platforms shall be of sufficient strength to support, two 275 pound firefighters, including full gear, on each of the side platforms and four 275 pound firefighters, including full gear, on the rear platform, without permanent deformation. In consideration of the truck departure angle requirements, the rear platform may fold up for storage. If a folding rear platform is used, it shall be lockable in both the up and down positions and the underside shall be covered with diamond tread plate.

3.6.6 Personnel steps. Personnel steps shall be provided to assist in gaining access to various firefighting components. Steps shall be provided on the forward end of the body for access to the transverse hosebeds, mounted at the right and left side of the truck at a height of not more than 17 inches above the side platforms. Steps shall be provided at the rear of the truck above the rear platform for access to the hose bed, mounted at a height of not more than 17 inches above the rear platform. Steps shall be commercially available or fabricated of diamond tread plate and shall be of sufficient size to accommodate one arctic or size twelve firefighter's boot.

3.6.7 Rear wheel housing. The rear wheel housing shall be circular type and large enough to permit the use of tire chains. The housing shall be fully splash tight to below the truck frame. Mud flaps shall be furnished at the rear of each wheel housing and shall be not less than the width of the tire(s). The rear wheel housing and surrounding area shall be designed to prevent damage to the body surface from tire chains.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

3.6.8 Handrails. The truck body shall be equipped with not less than one rear horizontal, two side and two rear vertical handrails. All handrails shall be of the same material. All handrails shall be readily accessible to personnel with physical dimensions in the range of small to large as defined in SAE J833. The rear horizontal handrail shall be not more than six inches forward of the rear of the hose bed, not more than 72 inches above the rear platform and the full width of the hose bed. Side vertical handrails shall be located to assist personnel climbing onto the side platforms and walkway. Rear vertical handrails shall be located at each side of the rear platform. Handrails shall not interfere with operations or removal of hose or equipment. One weatherproof push-button shall be installed on each of the rear vertical handrail supports and connected to a cab buzzer (see 3.4.11) or to the truck horn.

3.6.9 Equipment compartments. Equipment compartments shall be provided on each side of the truck fore, aft and above the rear wheelhousing, as space permits. Street side compartments shall extend to the top of the hosebeds and curb side compartments shall allow for mounting of the ladders. Side compartment tops shall be common, run continuously on each side of the body, include the compartment drip rail and be fabricated of diamond tread plate. The width, depth and height of the side compartments shall be the maximum practical. Street side compartments shall have one adjustable shelf for each 15 inches vertical height, while the curb side compartments shall have one adjustable shelf. All compartment doors shall be of roll-up design and gasketed to prevent entry of water when closed. Doors shall be equipped with automatic devices to hold the doors in any open position. Compartments shall be ventilated, have drain holes with rubber drain valves and Dri-Dec, or equal, on the floor. Adequate space shall be provided in a compartment for storage of one Self Contained Breathing Apparatus (SCBA). Individual air cylinder storage compartments, with doors and latches, for four one hour air cylinders shall be provided in otherwise unused spaces, such as between the wheelwell liner and adjacent compartments, below the cab floor, above the fire pump or adjacent to the booster hose compartment. A booster hose compartment shall be provided at the rear of the vehicle beneath the hose bed. All compartments shall be illuminated automatically by switches when the doors are opened.

3.6.10 Mounting brackets, holders and pads. Brackets, holders and pads, as applicable, shall be provided and installed on the truck for mounting fire fighting accessories listed in 3.6.11. Mounting attachments shall be designed to assure the equipment will remain in place under all truck operating conditions, yet allow quick release and removal of equipment for service. External mounting attachments shall be chromium plated brass or stainless steel, if commercially available, otherwise painted mounting attachments may be used. If ladders are not within reach of personnel with physical dimensions in the small range, as defined in SAE J833, while standing on the ground, a hydraulic assisted rack containing all of the ladders shall be used to lower and raise the ladders to and from the appropriate height.

3.6.11 Auxiliary fire fighting equipment. The following equipment shall be furnished with the truck. The equipment shall be located so that it is accessible and within reach from ground level.

a. Two 10 foot lengths of six inch hard suction hose conforming to ZZ-H-561, Grade B, Type I, with six inch NSFHT long handle female couplers on one end and six inch NSFHT long handle male couplers on the opposite end.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

Couplers shall be pyrolite or equal. The hoses shall be shipped loose with each truck.

b. One 20 foot length of five inch soft suction hose conforming to ZZ-H-451, Type III, Class B with six inch NSFHT long handle female gasketed coupling on one end and 4 1/2 inch NSFHT long handle female gasketed coupling on the other end. Couplings shall be pyrolite or equal. Hose must be approved for operating pressure of at least 300 PSIG.

c. Quick release fastening devices or rack shall be provided on the curbside of the truck, above the compartments, to mount the following complement of ladders. All ladders shall conform to NPFA Number 1931 and shall be of aluminum alloy.

(1) One 10 foot folding attic ladder.

(2) One 14 foot roof ladder with folding hooks.

(3) One 24 foot, three section tongue and groove constructed ladder with a closed length not to exceed 12 feet. If the ladder projects beyond the rear of the truck body, a protective cover shall be provided.

d. One eight foot fiberglass handle pike pole.

e. One six foot fiberglass handle pike pole.

f. One 5,000 watt continuous service, four cycle air cooled diesel engine driven, generator set with electric start capability. The generator set shall provide single phase 120/240V AC, 41.6/20.8 amp, 60 cycle power. The following output receptacles shall be provided: (2) 115V two prong polarized, one 115V twist lock and one 230V twist lock. The generator set shall be connected to the truck fuel tank with a quick disconnect fitting, and shall include an air cleaner, fuel filter with shutoff valve, low tone muffler, one gallon (minimum) accessory fuel tank, and an accessory rope starter. The generator set shall be installed in an equipment compartment with removable locking devices, on a slide out tray with locking devices to secure the tray in the stored and extended positions.

g. Two 500 Watt (110V AC) quartz or equivalent flood/spot lights with weatherproof Crouse-Hinds type connectors.

h. Two Electric (110/220V AC) smoke ejector fans with weatherproof motors and electrical connectors capable of 5000 CFM minimum air movement.

i. Two 100 foot extension cords, on reels, rated for use with portable generator, flood lights and smoke ejector fans listed above.

j. One flat head axe, six pounds, conforming to GGG-A-926, Type 1, Class 1

k. One pick head axe, six pounds, conforming to GGG-A-926.

l. One six inch female NSFHT, cylindrical suction strainer with lowering eye, constructed of Pyrolite, or equal. The lowering eye shall be not less than one inch inside diameter. The strainer screen shall be not greater than 1/4

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

inch mesh.

m. One swivel siamese suction fitting, identified as a six inch female NSFHT by (two) 2 1/2 inch gated female NSFHT with pin or rocker lug plugs constructed of pyrolite, or equal.

n. Six combination spanners, with hold downs, two on each side of truck at the suction discharge panels and two at the rear, conforming to GGG-W-665.

o. Two hydrant wrenches conforming to GGG-W-665.

p. One 2 1/2 female NSFHT by (two) 1 1/2 inch gated male NSFHT "Y" in accordance with MIL-C-52404.

q. One double female connection, 2 1/2 inch NSFHT in accordance with MIL-C-52404

r. One double male connection, 2 1/2 inch NSFHT in accordance with MIL-C-52404.

s. Two combination fog nozzles, 200 GPM.

t. Two combination fog nozzles, 95 GPM.

u. Space for one UL listed 20 BC rating dry chemical extinguisher.

v. Space for one UL listed 20 BC rating halon extinguisher.

w. Space for one UL listed 2A rating 2 1/2 gallon pressurized water extinguisher.

x. In addition to the above equipment, additional compartment space shall be provided for:

(1) Hooligan type tool.

(4) 12 foot by 12 foot salvage covers.

(1) 150 foot length of 1/2 inch fire service rope.

(4) Hose straps.

(1) Hose clamp.

3.6.12 Stored water system. The stored water system shall include water tank, piping and all required fill, drain and flushing components.

3.6.12.1 Water tank. The truck shall be equipped with a minimum 600 gallon capacity water tank. The water tank shall be constructed of passivated 316 stainless steel or fiberglass reinforced polyester. All materials used shall be compatible with water and water-foam solutions. The tank, with all covers in place, shall withstand an internal static air pressure of two PSIG for 30 minutes without leaking. Longitudinal and lateral baffles shall be incorporated to limit shifting of agent. The baffles shall be no more than 36 inches on

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

centers. The baffles and other tank interior structures shall be designed to allow access to all parts of the tank for cleaning and maintenance, and shall prevent vortices during pumping operations. The tank shall be provided with a, hinged or removable, gasketed inspection cover of at least 20 inches inside diameter. The tank shall be warranted against penetration from corrosion or deterioration for twelve years when used as specified. The water tank shall be mounted in such a manner to minimize stresses induced by frame flexing during operation.

3.6.12.2 Water tank vent. The water tank shall be provided with a combination overflow and vent of not less than three inches nominal pipe size extending below the truck frame and directed toward the ground.

3.6.12.3 Water tank fill and top fill. A female gated 2 1/2 inch NSFHT water tank fill connection, with chained on plug, shall be located on the curbside of the truck and shall allow the tank to be filled at a flow rate of 350 gallons per minute at 80 PSIG from a hydrant. The water tank shall be provided with a hinged and gasketed top fill cover with latch. The fill opening shall hold an unsupported 2 1/2 inch hose end, with male coupling, in place while the tank is filled at a flow rate of 350 GPM. The fill opening shall include a 1/4 inch mesh, nonferrous removable strainer. The tank top fill shall not be located in the hose bed area.

3.6.12.4 Water tank drain and discharge. The water tank discharge shall be located at the lowest point in the tank. The tank discharge (pump suction) line shall include a quick opening valve and a drain valve at the lowest point in the line. The drain valve shall allow complete draining of the water tank within five minutes. Valve controls shall be operable from the pump operator's control panel.

3.6.13 Booster hose and reel. The truck shall be equipped with an electric rewind booster hose reel centered in a compartment above the rear platform. Four 50 foot lengths of one inch inside diameter smooth bore rubber lined hose conforming to MIL-H-4497 shall be installed on the reel. A one inch combination fog and straight stream nozzle shall be connected to the end of the booster hose. The nozzle shall be securely mounted in a quick release device. A four way hose guide or roller shall be provided. The hose reel shall be connected to the pump discharge manifold with a minimum 1 1/2 inch diameter piping, having a flexible section to take up vibration, and shall incorporate an air blow-down, controlled from the booster hose compartment. The hose reel shall have a removable hand crank rewind and shall be equipped with a friction brake payout to prevent overtravel or the reel from turning due to pressure surges.

3.6.14 Foam equipment. The truck shall be equipped with a foam system as specified herein. The foam system shall be suitable for operation with AFFF, conforming to MIL-F-24385. The system shall be capable of discharging a three percent foam solution at all pump discharge outlets.

3.6.14.1 Foam proportioner. The foam proportioner shall be of the around the pump design. The AFFF concentrate shall be inducted by venturi action into a line carrying a portion of the agent from the fire pump discharge routed into the pump intake and shall mix with the main water flow. An adjustable metering valve shall be provided in the foam liquid line leading into the venturi for controlling the amount of foam liquid entering the venturi.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the current version before use.

3.6.14.2 Foam controls. All gauges and controls for the operation of the foam system shall be plated or polished and grouped together on the fire pump operator's control panel. The foam metering valve control shall be graduated for total GPM discharge.

3.6.14.3 Foam liquid tank. A removable, minimum 55 gallon capacity, foam liquid tank shall be provided. The foam liquid tank shall be constructed of passivated 316 stainless steel or fiberglass reinforced polyester. A 1 1/2 inch drain, with NSFHT threaded end, controlled from the pump operator's panel, shall be provided. The drain shall discharge at a point below the bottom of the truck frame. The tank shall have a filler opening, adequate for filling directly from five gallon containers, which will not interfere with other firefighting equipment operation and shall be provided with a tight gasketed hinged cap with a spring type hold down latch. A removable strainer of not greater than number 10 mesh shall be provided in the fill opening. The foam tank outlet shall be designed to prevent vortexing. The foam tank shall be provided with at least a one inch vent that will not permit spillage of foam when the truck is operated on the specified grades and side slopes and when cornering the truck. The tank shall be warranted against complete penetration from corrosion or deterioration for twelve years when used as specified.

3.6.14.4 Foam system piping. All fittings, valves and piping shall be of materials that are fully compatible with the AFFF concentrate, water and their admixtures. Welded connections shall be used to the maximum extent possible. Quarter turn ball valves shall be installed between the foam tank and the metering valve and between the fire pump discharge and the eductor. A ball check valve shall be installed between the eductor and metering valve to prevent flow of water into the foam tank. A horizontal swing check valve shall be installed in the water tank suction line to prevent entry of foam solution into the water tank.

3.7 Low temperature provisions.

3.7.1 Winterization for -20°F operation. The truck shall be capable of operating in ambient temperatures as low as -40°F. The truck engine shall start in ambient temperatures of -20°F, and shall be equipped with thermostatically controlled radiator shutters and/or a thermostatically controlled engine fan, an ether injection system and other provisions required to meet this performance. All trucks shall be so designed that the additional features required for winterization to -40°F, as described in 3.7.2, may be readily installed. To meet the -20°F operational requirements, at least the following minimum provisions shall be incorporated into the truck:

a. A fully enclosed and insulated fire pump compartment with removable access panels.

b. A minimum one inch thick insulation between the water tank and the surrounding truck body and hose bed floor.

c. An insulated hose reel compartment.

d. Insulation shall be of expanded polyethylchloride or polyvinylchloride materials and shall not absorb water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.7.2 Winterization for -40°F operation. When specified (see 6.2), a winterization system shall be provided that will permit satisfactory operation of the truck in ambient temperatures as low as -40°F. The winterization system shall be powered by the truck electrical system, with the engine running at idle speeds or with the engine shut off and the batteries being maintained by the battery charger while connected to an external 110V AC power source. Automatic switches shall be provided to shutoff the system in the event of starting or overheating failures. The on-off control switch and a green light to indicate when the system is operating shall be installed on the cab instrument panel. The system shall be so designed and installed that major components of the system can be removed and reinstalled without draining coolant from the system and without draining of coolant in excess of the capacity of the component being removed. To meet the -40°F operational requirements, at least the following minimum provisions shall be incorporated in the truck:

a. A combustion, liquid coolant circulating or hot air heating, automatic cycling type heater for heating the agent tanks, the fire pump compartment and the hose reel.

3.8 Part numbering and interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.9 Dimensions. The truck shall conform to the following dimensions:

a. Width - 96 inches maximum.

b. Height - The overall height of the truck in a fully operational configuration, except for water, shall not exceed 118 inches.

c. Length - Minimum practical.

d. Driver's Compartment - The distance from the centerline of the front axle to the rear of the cab shall not exceed 30 inches.

e. Angles of approach and departure of not less than 17° (4X2) and 25o (4X4).

3.10 Finishes and protective coatings.

3.10.1 Cleaning, painting, plating, anodic films and chemical treatments. Surface preparation, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808. Primers shall be in accordance with MIL-P-23377 and MIL-P-26915 for aluminum and ferrous alloys respectively.

3.10.1.1 Finishes. Unless otherwise specified, all metallic surfaces, both interior and exterior, shall be finished and treated for corrosion prevention. Priming shall be the minimum finish requirement. All faying surfaces shall be primed and completely coated with sealant before assembly. All mechanically joined components shall be finished before assembly. The paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion,or other defects and shall pass the wet tape test described in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

4.6.20 of this specification. A corrosion prevention and control plan shall be included in the technical proposal describing the contractor's plan for surface preparation and finish for all materials in accordance with MIL-STD-1568, paragraphs 5.1.1 and 5.1.2. The corrosion prevention and control plan does not constitute release from any specification requirements.

3.10.2 Topcoat.

3.10.2.1 Lime Yellow. Cab and body exterior surfaces and wheels shall be painted with MIL-C-83286, Lime Yellow, Color Number 13670 of FED-STD-595. Compartment doors and interior surfaces, all diamond tread plates, walkways, steps, side and rear platforms and protective panels shall not be painted. The radiator and grille shall be chromium plated or polished stainless steel. The front bumper shall be chromium plated or polished stainless steel.

3.10.2.2 Forest green. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments, except in the cab, shall be painted with MIL-C-83286, Forest Green, Color Number 24052 of FED-STD-595. All exterior trim items, such as wheels, wheel covers, running boards, metal radiator grilles, bumpers, light housings, trim rings, door handles and drip rails, shall be the same color as the exterior finish of the vehicle.

3.10.2.3 Desert sand. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments, except in the cab, shall be painted with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items, such as wheels, wheel covers, running boards, metal radiator grilles, bumpers, light housings, trim rings, door handles and drip rails, shall be the same as the exterior color of the vehicle.

3.10.3 Cab interior. Cab interior metal surfaces shall be wrinkle finish low luster black paint, black vinyl covered or a combination of both.

3.11 Rustproofing. The truck shall be rustproofed and undercoated in accordance with FED-STD-297D and shall be applied only after the finish coating has been applied. The level of corrosion treatment shall be tropical. The use of proprietary materials shall require Government approval. The first article trucks shall not be rustproofed or undercoated in visually accessible areas until after first article testing.

3.12 Marking. Markings for trucks painted Lime Yellow shall have reflectorized tape applied in accordance with L-S-300, Type 1, Class 3, Reflectivity 1, white on a red background. Markings for trucks painted Forest green shall be applied with black paint, Color Number 37038 of FED-STD-595. Markings for trucks painted Desert sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.12.1 Lettering and numbering. All lettering and numbering shall be in accordance with Technical Order T.O. 36-1-3.

3.12.2 Tire pressure. The recommended tire inflation pressure shall be stenciled on the truck as near to each tire as practical and on the vehicle data plate (see paragraph 3.13).

3.12.3 Reflective band. Trucks painted Lime Yellow shall have a horizontal 10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

inch wide white reflective band applied to the painted surfaces of the cab and truck body at a level near the bottom of the windshield. The band shall be interrupted at unpainted surfaces of tread plate and compartment doors.

3.13 Data plates. Instructions and identification data shall be permanently and legibly etched upon metal plates in accordance with MIL-P-514. All plates shall be of a size and shape consistent with information required thereon. Abbreviations shall be in accordance with MIL-STD-12. Symbols shall be in accordance with MIL-STD-17. Data plates shall be attached to the truck with rivets.

3.13.1 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. A nameplate shall be mounted in the cab with screws or rivets, and except for serial number and date of delivery which may be stamped, all data shall be inscribed. The following nomenclature shall be added to the nameplate:

Truck, Fire Fighting, 1,250 GPM, Four by Two or 1,250 GPM, Four by Four

Water Capacity	Gallons
Vehicle Weight Unloaded	Pounds
Gross Vehicle Weight	Pounds
Fuel Grade	
Oil, Engine, Above 32°F, SAE Grade	
Oil, Engine, Below 32°F, SAE Grade	
Date of Delivery	

3.13.2 Identification of controls. All valves, gauges, controls, and indicators, other than standard instrumentation used in operation of the truck, shall be identified by nameplates, screwed or riveted to the truck, and of such composition that exposure to oil, dirt, light, etc., will not cause them to fade or become eradicated. Tags and decals shall not be used.

3.13.3 Operating instructions. Brief operating and precautionary instructions shall be screwed or riveted near the appropriate control or switch and shall be clear, concise, and adequate to enable operation without damage to the equipment or injury to personnel.

3.13.4 Caution plate. The following caution plate shall be screwed or riveted at a conspicuous location on the cab dash panel.

CAUTION

THIS TRUCK IS EQUIPPED WITH RADIO INTERFERENCE SUPPRESSION DEVICES. CARE SHOULD BE EXERCISED DURING TRUCK MAINTENANCE TO RETAIN AND REINSTALL THESE DEVICES PRIOR TO REUSE OF THE TRUCK.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

3.13.5 Flushing instruction plate. An instruction plate shall be screwed or riveted on the pump control panel and shall contain specific data concerning flushing of the agent system piping.

3.13.6 Lubrication instruction plate. A lubrication plate shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for operational temperatures. The plate shall be screwed or riveted to the truck in an accessible location.

3.13.7 Pump delivery rate. A plate shall be screwed or riveted on the pump control panel indicating the fire pump water delivery rate in GPM at 150, 200 and 250 PSI pressures with the truck manufacturer's specified engine speeds for these delivery rates.

3.13.8 Transportation data plate. A transportation data plate shall be screwed or riveted to the truck body and shall contain at least the following:

a. Side and rear silhouette views of the truck.

b. Horizontal and vertical location of the truck center-of-gravity marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width and height.

3.14 Workmanship. The truck, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges, marking of parts and assemblies, separation of dissimilar metals, thoroughness of welding, riveting, painting, and wiring, alignment of parts and torque of assembly fasteners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of tests. The inspection and testing of the truck shall be classified as follows:

a. Preproduction testing (see 4.4).

b. Acceptance tests (see 4.5).

4.3 Test conditions.

4.3.1 Preparation for test. All trucks submitted for test shall contain all specified components and shall be completely assembled and properly prepared for operation. The truck, including all components, shall be correctly adjusted, lubricated, and serviced and all equipment which has been ordered shall be mounted for immediate operation. Fuel and water tanks shall be full unless otherwise specified for a particular test. Fuel to be used for all testing shall be Grade JP-8 in accordance with MIL-T-83133.

4.3.2 Data. During the tests specified herein, at least the following data, as applicable, shall be recorded at intervals not greater than 15 minutes or as otherwise indicated. Additional data may be required and recording intervals decreased for tests of short duration or as otherwise dictated by test objectives.

a. Time started and completed.

b. Ambient temperature and atmospheric pressure.

c. Level in agent tanks.

d. Fire pump inlet pressure.

e. Fire pump discharge pressure.

f. Agent flow rate through each hose line.

g. Engine speed (RPM).

h. Engine oil pressure.

i. Engine coolant temperature.

4.3.3 Test observations. Throughout all tests specified herein, the truck shall be observed for the following conditions which shall be considered cause for rejection. If a failure occurs, a critical mechanical inspection shall be made of all related components to determine their operability, any damage or undue wear incurred during testing. Teardown and parts measurement shall be made in those cases where service life is in question.

a. Failure to conform to design and performance requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

b. Spillage or leakage of any liquids including fire fighting agent, coolants, fuels, and lubricants under all operating conditions.

c. Structural failure of any part including permanent distortion, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the truck and the ground or between truck components or all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of parts and components.

g. Instability in agent pressure and flow.

h. Evidence of instability in handling during braking, turning and negotiating slopes.

i. Evidence of a safety hazard to personnel during operation, service and maintenance.

j. Overheating of the truck engine or any component of the truck.

4.4 Preproduction testing (see 6.2).

4.4.1 Test sample. One (4X4) type truck shall be subjected to the tests shown in 4.4.3. Preproduction tests. One of each type truck specified shall be subjected to the tests shown in 4.5.1 Individual tests and 4.6.19 Operational tests.

4.4.2 Test Report. After the completion of the preproduction tests, a test report for each type of truck shall be prepared in accordance with MIL-STD-831.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests (see 4.5.1).

b. Random sample testings (see 4.5.2).

4.5.1 Individual tests. Each truck shall be subjected to the following tests as described under 4.6:

a. Examination of product (see 4.6.1).

b. Agent tanks, static pressure test (see 4.6.2).

c. Agent system, hydrostatic pressure test (see 4.6.3).

d. Twenty five mile road test (see 4.6.4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

e. Pumping tests (see 4.6.11.1 through 4.6.11.3).

f. Paint adhesion tape test (see 4.6.20).

4.5.2 Random sample testing. One truck may be selected by the procuring activity at random from each fifty or fraction thereof produced and subjected to the following tests as described under 4.6. Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets of the report.

a. One hundred mile road test (see 4.6.4.1).

b. High temperature pumping test (see 4.6.11.4).

c. Agent discharge test (see 4.6.12).

d. Low temperature operation (see 4.6.14).

4.5.2.1 Rejection and retest. When a truck selected from a production run fails to meet the specification, no trucks still on hand or later produced shall be accepted until the extent and cause of failure has been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated at the contractor's expense.

4.5.2.2 Individual tests may continue. Individual tests may be continued pending the investigation of a test failure. Final acceptance of the trucks on hand or produced later shall not be made until it is determined that all trucks on hand have been corrected.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of defects likely to be found, methods of correction and disposition of parts required.

4.6 Test methods.

4.6.1 Examination of product. The truck shall be inspected to determine compliance with requirements of this specification with respect to materials, dimensions, workmanship, and markings. A thorough inspection shall be made of to insure proper functioning of all mechanical and electrical components, accessibility of all controls and levers, and safe operation. All controls shall be moved through their maximum normal displacement operations and shall exhibit no malfunctions, leaks, rough operation or irregular movements.

4.6.2 Agent tanks static pressure test. The agent tanks shall be subjected to an internal static air pressure of two PSI for not less than 30 minutes. A soap solution shall be sprayed on the exterior of the tank to facilitate detection of leaks that may exist. This test may be performed with the tank separate from the truck.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

4.6.3 Water system hydrostatic pressure test. The water system piping, including all components subject to the discharge pressures of the pump shall be subjected to a hydrostatic pressure of 350 PSI for a period of 10 minutes with no loss of pressure.

4.6.4 Road tests. Unless otherwise noted, refer to Paragraph 4.6.16.10 Endurance for distance requirements of road testing.

4.6.4.1 Paved road test. The truck shall be operated over hard surface roads for the distance specified at an average speed of 45 MPH. During the test, the following functions shall be performed for each 100 mile portion:

a. Normal or average stops, ten times.

b. Quick stops from a speed of at least 20 MPH by instantaneous application of full service braking power, five times.

c. Sharp right and sharp left turns, five times each.

d. Operation through all forward and reverse gears.

e. Maintain a speed of 55 MPH up a three percent grade not less than 1/2 mile long.

4.6.4.2 Twenty five mile road test. The fully loaded truck shall be operated over hard surface roads for a distance of 25 miles at a minimum average speed of 35 MPH. During this test, the same tests specified in a. through e. of paragraph 4.6.4.1 Paved road test shall be performed.

4.6.4.3 Unimproved road test. The truck shall be operated over unimproved surface gravel roads for the distance specified at an average speed of 25 MPH. During this test, the same tests specified in a. through d. of paragraph 4.6.4.1 Paved road test shall be performed.

4.6.4.4 Hilly cross country test. (4X4 truck only). The truck shall be operated over hilly cross country terrain for the distance specified at speeds up to 15 MPH. During the test, the following functions shall be performed for each 25 mile portion:

a. Normal or average stops, ten times

b. 50 percent grade stops and starts, headed up and headed down, five times each.

c. Operation through all forward and reverse gears five times.

d. Operations of the axle differential locking controls for a minimum of one mile.

e. 20 percent side slope operation with turns up and down the slope, five times each.

4.6.5 Ramp test. The fully loaded truck, except for draining of agent tanks, shall be driven forward from a solid horizontal surface and ascend a rigid ramp

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

equal in length to at least 1.2 times the truck wheelbase and making an angle of 17° with the horizontal. The truck shall proceed across the ramp crest onto an upper connecting horizontal surface until all wheels are resting on the upper surface. The truck shall then be backed from the upper horizontal surface, down the ramp and onto the lower horizontal surface. The truck shall then be turned around and backed up the ramp onto the upper horizontal surface and then driven forward down the ramp and onto the lower horizontal surface.

4.6.6 Pump removal. It shall be demonstrated that the fire pump can be removed from the pump compartment within the allowable service time limit without requiring removal of other major components.

4.6.7 Top surface loading. It shall be demonstrated that the top surface of the truck body, extending from the forward bulkhead to the rear of the hose bed, will withstand a load of 200 pounds per square foot at any location without permanent distortion, damage or evidence of lack of rigidity.

4.6.8 Platform loading. It shall be demonstrated that the rear platform will support four 275 pound firefighters, including gear, and the side platforms will each support two 275 pound firefighters, including gear, without damage or permanent deflection when used in the environment the individual truck are designed for.

4.6.9 Water tank capacities. The agent tanks shall be filled to capacity and the quantity required shall be accurately measured. It shall be demonstrated that the water tank can be filled from any of the 2 1/2 inch suction connections on the truck and from the tank top fill using 2 1/2 inch hose at a fill rate of 350 GPM at 80 PSIG.

4.6.10 Hose reel operation. It shall be demonstrated that the booster hose can be easily withdrawn from the reel at various angles away from the truck without operational interference. The hand crank rewind and the hose reel friction brake shall be demonstrated.

4.6.11 Pumping tests. The following pumping tests shall be conducted and recorded in accordance with NFPA 1901 and as specified herein.

4.6.11.1 Run in test. The truck shall be subjected to a run in test of not less than three continuous hours while drafting water at a 10 foot lift through 20 feet of five inch hard suction hose. The fire pump shall be completely drained at the beginning of this test, and the time for the primer pump to initiate flow shall be recorded. During this test, the fire pump shall discharge agent as follows:

a. 1250 GPM at 150 PSI for 2 hours continuous pumping.

b. 1250 GPM at 165 PSI for 10 minutes continuous pumping.

c. 875 GPM at 200 PSI for 1/2 hour continuous pumping.

d. 625 GPM at 250 PSI for 1/2 hour continuous pumping.

4.6.11.2 Pump vacuum test. The pump interior, with capped suctions and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

uncapped discharges shall be tested to a vacuum of 22 inches of Hg. by means of the truck priming pump and shall not drop more than 10 inches in 10 minutes. The priming pump may not be used after the test has begun. The vacuum may be reduced by one inch for each additional 1000 feet altitude above 1000 feet.

4.6.11.3 Pressure control test. With the fire pump operating at maximum rated discharge capacity for the respective pressure and the pressure relief valve set at 10 PSI higher than the fire pump working pressure, all discharge outlets shall be slowly closed 90, 150, 200 and 250 PSI, and the discharge pressure increase recorded under each condition. Pressure increases shall not be greater than 30 PSI.

4.6.11.4 High temperature pumping test. The truck shall be placed in a hot chamber and subjected to an ambient temperature of 125°F, plus or minus 2°F, for a period of at least 24 hours. Upon completion of the 24 hour period and while maintaining the 125°F temperature, the truck engine shall be started and the pump operated for three continuous hours while drafting water through 20 feet of five inch hard suction hose. The fire pump shall be completely drained at the beginning of this test, and the time for the primer pump to initiate flow shall be recorded. During this test, truck engine and fire pump temperatures shall be monitored. The fire pump shall discharge water as follows:

a. 1,250 GPM at 150 PSI for two hours continuous pumping.

b. 875 GPM at 200 PSI for 1/2 hour continuous pumping.

c. 625 GPM at 250 PSI for 1/2 hour continuous pumping.

The air conditioner shall be used at this time to prove conformance to paragraph 3.4.5.2.

4.6.11.5 Gear lubricant temperature test. During the high temperature pumping test, the temperature of the gear lubricant shall be recorded and shall not exceed the maximum allowable operating temperature recommended by the pump manufacturer for continuous operation.

4.6.12 Agent discharge test. Agent shall be discharged from the truck to determine that:

a. Water discharge rates of 750 GPM can be obtained through any two of the 2 1/2 inch discharge outlets while pumping water at a rate of 1,250 GPM from draft or hydrant supply.

b. A water discharge rate of 200 GPM through any of the 1 1/2 inch preconnect discharge outlets separately or simultaneously while pumping from draft, hydrant or the installed water tank supply.

c. A water discharge rate of 60 GPM can be obtained through the booster hose while pumping from draft, hydrant or the installed water tank supply.

d. A water discharge rate of 60 GPM at a maximum pump pressure of 150 PSI during continuous forward motion or stop/start operation of a fully loaded truck from level hard surface or while ascending a 50 percent grade at speeds not to exceed 5 MPH. (4X4 trucks only).

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:47Z
Check the source to verify that this is the current version before use.

e. The monitor (deluge gun) shall be tested to verify a minimum water flow rate of at least 1,250 GPM from draft or hydrant supply. The maximum monitor flow rate from draft shall be recorded.

f. The foam system shall be tested using AFFF conforming to MIL-F-24385. Foam solution shall be operated for not less than one minute while discharging 250 GPM through one of the 2 1/2 inch discharges with 100 feet of 2 1/2 inch hose and fog nozzle. The system shall use not less than 7.5 gallons of foam liquid per minute, plus or minus five percent. The monitor shall be tested with AFFF for not less than one minute at 1000 GPM. The system shall use not less than 30 gallons of foam liquid per minute, plus or minus five percent.

4.6.13 Water tank exhaustion. It shall be demonstrated that at least 90 percent of the water tank capacity can be discharged during pumping operations while the truck is either headed up or headed down on 30 percent slope (4X2) or on 50 percent slope (4X4), or while positioned in right or left directions on a 20 percent side slope.

4.6.14 Low temperature operation. The fully winterized truck shall be serviced with low temperature lubricants and coolants. The shall be instrumented and shall record at least the truck engine, fire pump compartment and water tank temperatures. The truck shall be placed in a cold room and subjected to the following conditions and tests:

a. The truck, with dry fire pump and empty water tank, shall be subjected to an ambient temperature of -20°F, plus or minus 2°F, for a period of at least eight hours. At the end of this period, it shall be demonstrated that the truck engine can be started without assist from winterization system.

b. Upon completion of the engine start test, the water tank shall be filled to capacity. Temperatures of the water used in filling the tank shall not exceed 60°F. With the water tank filled, the truck shall continue to be subjected to an ambient temperature of -20°F, plus or minus 2°F, for a period of at least two hours. At the conclusion of the two hour period, the fire pump shall be operated and the capability to discharge water shall be demonstrated by operating each discharge outlet and the booster hose. There shall be no evidence of freezing in the system.

c. Upon completion of the -20°F test, the winterization system shall be activated, the water tank filled and the fire pump flooded, the chamber temperature shall be reduced to -40°F, plus or minus 2°F, for a period of at least four hours. At the conclusion of the four hour soak period, the engine shall be started in accordance with 3.3.1.3, the fire pump shall be operated and the capability to discharge water shall be demonstrated by operating all discharge outlets and the booster hose. It shall be demonstrated that the booster hose can be withdrawn and rewound on the reel by one firefighter. There shall be no evidence of freezing in the water system.

d. The heater and defroster shall be used at this time to indicate conformance to paragraph 3.4.5.1.

4.6.15 Electromagnetic interference. The truck shall be subjected to the radio interference suppression tests in accordance with Notice 3, Class D3, Test

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

Method RE05 of MIL-STD-461 with any and all electrically activated equipment used in or on the truck while operating (see 3.1.15).

4.6.16 Truck mobility and performance tests.

4.6.16.1 Dimensions. Measurements of the truck overall length, width, height, wheelbase, approach and departure angles, interaxle clearances, wall to wall turning circle and rear of cab to centerline of front axle shall be made.

4.6.16.2 Weight. The curb and gross weights of the truck shall be determined. (see 3.1.12).

4.6.16.3 Load distribution. The load on each wheel and on each axle of the truck at curb and gross weights shall be determined (see 3.1.12.3).

4.6.16.4 Braking. The fully loaded truck, including firefighters, shall be brought to a stop on a dry hard surface, free of loose material, using the service brakes when headed both up and down a 30 percent grade (4X2), or a 50 percent grade (4X4). The parking brake shall be applied, the service brakes released and the truck shall remain stationary. The truck shall be driven at a speed of 20 MPH on a level hard surface, free of loose material, the service brakes shall be applied and the distance required to bring the truck to a stop shall be measured (see 3.2.1(g)).

4.6.16.5 Maximum speed. The fully loaded truck, including firefighters, shall be operated on a level paved course to determined the maximum vehicle speed that can be obtained (see 3.2.1(b)). The truck shall also demonstrate the capability to maintain a speed of 55 MPH up a three percent grade (see 3.2.1(o)).

4.6.16.6 Acceleration. The fully loaded truck, including firefighters, shall be accelerated from a standstill to 35 MPH on a level paved course within 20 seconds (see 3.2.1(c)).

4.6.16.7 Gradeability. The fully loaded truck, including firefighters, shall be driven up and down a 30 percent grade (4X2), 50 percent grade (4X4). The ability of the truck to negotiate this grade and to stop and start while both ascending and descending shall be demonstrated (see 3.2.1(d)). Engine oil pressure shall be recorded during grade operations. There shall be no evidence of loss of fuel, lubricants or fire fighting agents.

4.6.16.8 Side slope. The fully loaded truck, including firefighters, shall be driven in both directions on a 20 percent side slope and shall demonstrate turns both up and down the slope. The ability to stop and start the truck in both directions on the side slope shall be demonstrated (see 3.2.1(e)). Engine oil pressure shall be recorded during grade operations. There shall be no evidence of loss of fuel, lubricants or fire fighting agents.

4.6.16.9 Endurance. The truck shall be operated under the conditions listed and over the following roads and distances specified. After each segment of testing the agent pump shall be operated at:

1250 GPM at 150 PSIG for 30 minutes,

875 GPM at 200 PSIG for 15 minutes, and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

625 GPM at 250 PSIG for 15 minutes.

Leaks from the pumping system, in excess of that allowed by NFPA 1901, shall be cause for rejection of the entire Endurance test completed up to the time of failure.

a. Paved roads at an average speed of 45 MPH for 200 miles fully loaded, plus 200 miles with water tank filled to one half capacity, plus 200 miles unloaded except for driver. During the paved road operation, the truck shall be accelerated from a standstill to its maximum rated speed after each 200 miles. At the completion of the paved road operation, the engine shall be operated at idle for four hours. The truck shall be evaluated for faulty operation, leaks, missalignment, and damage throughout the test. The distance traveled to the test site, if required, may be used to satisfy a portion of this test phase providing complete records are kept during the trip.

b. Graded gravel roads at an average speed of 25 MPH for 100 miles fully loaded, plus 100 miles with water tank filled to one half capacity, plus 100 miles unloaded except for driver.

c. Hilly cross country (4X4 truck only) at speed up to 15 MPH for 100 miles fully loaded, plus 100 miles with water tank filled to one half capacity, plus 100 miles unloaded, except for driver.

4.6.16.10 Auxiliary generator test. A test shall be conducted to verify the auxiliary generator output capacities (see 3.6.11(f)).

4.6.17 Sound level testing. The vehicle shall be tested in accordance with SAL J336a and J366 to demonstrate compliance with 3.1.16.

4.6.18 Reliability and maintainability demonstration.

4.6.18.1 Reliability demonstration. Compliance with the quantitative reliability requirements of 3.1.3.2 shall be demonstrated by all tests specified herein. Apparent failures that are caused by some unintended factor external to the equipment (e.g.; test equipment failure) shall not be counted as reliability failures if the cause is justified to the procuring activity. Any test resulting in a failure shall be fully documented. Servicing, preventive maintenance or adjustments required by the technical manuals shall not be counted as failures.

4.6.18.2 Maintainability demonstration. Demonstration of maintenance task performance requirements of 3.1.3.3 and 3.1.3.3.1 shall be considered as compliance with the maintainability downtime requirements. Only active maintenance time shall be counted. Maintenance tasks shall be selected by the procuring activity from a list of possible maintenance tasks and estimated performance times submitted by the contractor. Not less than 25 percent or more than 50 percent of the total number of tasks listed shall be demonstrated to verify an average maintenance downtime. Performance times shall be adjusted if tasks are demonstrated by personnel possessing higher skills than those required for the technical manuals. The capability to perform servicing and maintenance tasks by personnel wearing heavy gloves or arctic mittens shall be demonstrated.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:47Z
Check the source to verify that this is the currentversion before use.

4.6.19 Operational test. The contractor shall provide one each (4X2) and (4X4) trucks for operational testing. Operational test vehicles shall not be the same vehicles used for preproduction testing. Each vehicle shall be shipped to a Government designated site for purposes of estimating for 60 days of operational use concurrent with preproduction testing. The test period shall begin after each vehicle has been inspected per paragraphs 4.6.1 and 4.6.2 and prepared for use. The trucks shall be available for any 12 hour period each day, seven days a week, for a total of 720 hours. The contractor shall provide one person to accompany each truck, during all operations, to instruct in the use of the controls and to record vehicle failures. The vehicles shall be inspected, serviced and operated by Government personnel using the procedures contained in the preliminary technical data supplied with the trucks. Fuels, lubricants, water and foam agent shall be provided by the Government for use during the operational testing. During each day of operation, up to 15 pan fires or simulated structural fires may be extinquished. The vehicles shall be driven up to 200 miles per day, over hard surfaces, gravel roads and cross country terrain (4X4 truck only). Sudden stops, minimum radius turns, maximum speed and acceleration may be included in daily operations. All systems and subsystems of the vehicles may be tested at the beginning and ending of the test period. The Government may perform any test to demonstrate compliance with this specification. Any failure or condition detrimental to the operation shall constitute a reliability failure. Any period of more than four hours that the vehicles are out of service for maintenance or repair shall not be considered part of the test period. Any parts or corrective actions required shall be provided by the contractor. As many as twenty minor adjustments per truck are allowed during the test period. The contractor shall be responsible for recording all operations, servicing and corrective actions required for the first article test report.

4.6.19.1 Reliability and maintainability information. The following information shall be an attachment accompanying the operational test report or shall be included as part of the report:

a. All failures, maintenance and other events recorded shall be identified by accumulated operating time, miles, cycles or position in the test procedure as appropriate. Test conditions during the failures shall be recorded.

b. Description of the engineering reasoning or tests conducted to determine assignable cause for all failures and irregular operations identified.

c. Description of the engineering reasoning behind any corrections made or proposed to be made, and the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that should improve the item.

e. Test activity or contractor comments on field conditions or procedures to be avoided that may increase the truck's reliability and useful life.

f. Record of man-hours required for each maintenance and servicing action during the tests. A brief description of the qualifications and experience of the personnel involved shall be included and shall be adequate to allow comparison to the personnel anticipated in similar field work.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:47Z
Check the source to verify that this is the currentversion before use.

4.6.20 Paint adhesion (wet) tape test. This test establishes a minimum acceptability of intercoat and surface adhesion of polyurethane coating systems. This test shall be applied to the completed exterior finish of the vehicle after drying for a minimum period of 48 hours. Only one area shall be tested unless failure occurs, then not less than five additional tests shall be performed. Three consecutive failures shall constitute rejection. Failures are defined as visible peeling or chipping of the painted surface. Perform the wet tape test as follows:

a. Soak a clean cloth in distilled water and fold into a three inch square at least four layers thick.

b. Place the cloth on the area to be tested and cover with a piece of waterproof plastic. Tape the plastic sheet to the painted surface, completely sealing the edges to prevent evaporation of the water in the cloth. Allow the wet cloth to contact the test area for a minimum of 24 hours.

c. Remove the plastic and wet cloth and blot up the wet surface. Immediately apply a one inch wide strip of tape, (Code No. 250 3M Company, age of tape shall not exceed one year), adhesive side to the painted surface. Press the tape down with two passes of a 4 1/2 pound rubber covered roller.

d. Remove the tape in one abrupt motion and examine the tested area for damage. Removal of one of the layers of the paint system or the paint system from the metal shall constitute failure.

4.6.21 Water tightness. The truck shall be subjected to Method 506.2, Procedure I of MIL-STD-810 to assure water tightness of the cab. Any leakage into the cab shall be cause for rejection.

4.7 Refurbishment. After successful completion of all required testing, each vehicle shall be restored to like new condition, except for accumulated mileage.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The truck is intended to be used as a structural and brush-structural firefighting vehicles for use at military installations world wide for protection of real property and material.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type vehicles to be furnished (see 1.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

c. Issue of the DODISS to be cited in the solicitation and, if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).

d. When performance requirements shall be based on operation above 3,000 feet above sea level (see 3.1.19).

e. When engine braking is required (see 3.3.1.5).

f. When winterization for -40°F operation is to be provided (see 3.7.2).

g. Finish color required (see 3.10).

h. Lettering and numbering required on the truck (see 3.12).

i. Location and conditions for preproduction testing (see 4.4).

j. When blue colored light bar, warning lights and forward flashing lights are required (see 3.3.9.4.4).

k. When heated rearview mirror are required (see 3.4.6).

6.3 Definitions.

6.3.1 Automatic transmission. An automatic transmission is one in which all combinations of variable torque ratio changes can be accomplished by the operator with one setting of the shift control and without more than an instantaneous interruption of the power flow.

6.3.2 Cross country terrain. Cross country terrain consists of open fields, broken ground and uneven terrain in general.

6.4 Subject term (key word) listing.

Fire truck
Structural
Brush
Fire Hose
Pump

Custodian:
Air Force - 99

Preparing activity:
Air Force - 84

Project 4210-F422

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-83823	13 Sep 91

3. DOCUMENT TITLE
TRUCK, FIRE FIGHTING (STRUCTURAL OR BRUSH-STRUCTURAL, 1,250 GPM, 4 X 2 OR 4 X 4 CHASSIS)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) WR-ALC/LZDTB Robins AFB GA 31098-5365	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 20260-26104:47Z
Check the source to verify that this is the currenversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:47Z
Check the source to verify that this is the current version before use.